Chapter 3: The Link Layer

Our goals:

- understand principles behind data link layer services:了解数据链路层服务的原理：
  - framing成帧
  - error detection, correction错误检测、纠正
  - reliable data transfer, flow control可靠的数据传输、流量控制
  - sharing a broadcast channel: multiple access多路访问
  - link layer addressing链路层寻址

Overview:

- link layer services
- framing
- error detection, correction
- reliable data transfer
- multiple access protocols and LANs局域网
- link layer addressing

Keypoints and Difficulties

Keypoints:

- Four framing methods四种成帧方式
- CRC CRC校验码
- reliable data transfer可靠交付
- multiple access protocols
- link layer addressing

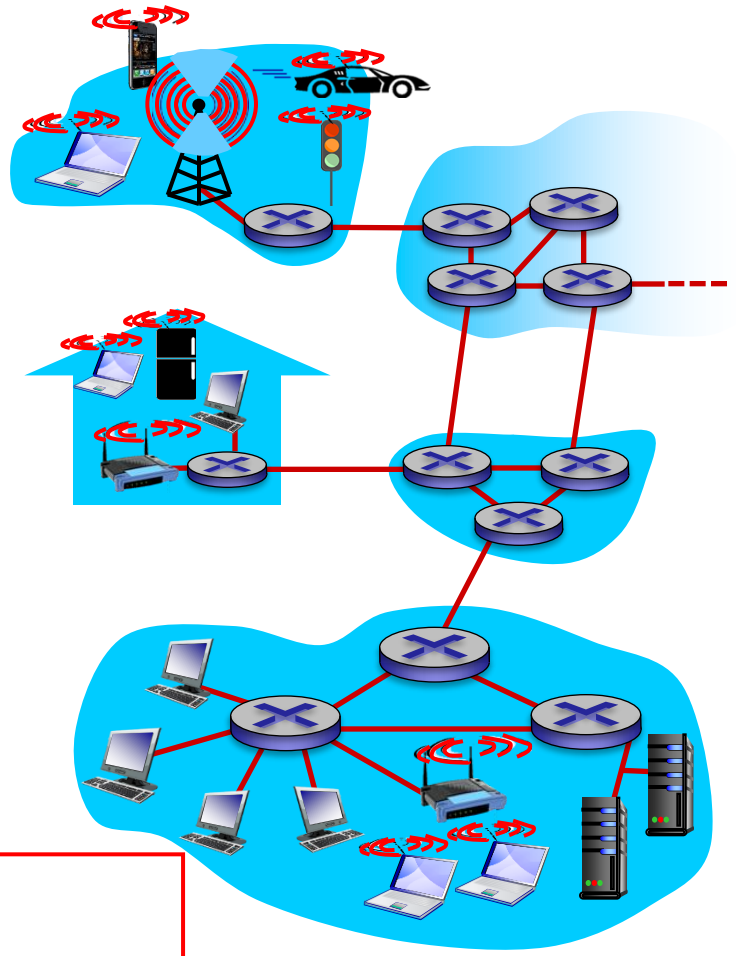
Difficulties:

- CRC
- Flow control
- Sliding-window protocol
- CSMA/CD

Link layer: Introduction

terminology:

- hosts and routers: nodes
- communication channels that connect adjacent nodes along communication path: links
  - wired links
  - wireless links
  - LANs
- layer-2 packet: frame, encapsulates datagram



data-link layer has responsibility of transferring datagram from one node to *physically adjacent* 邻近的 node over a link

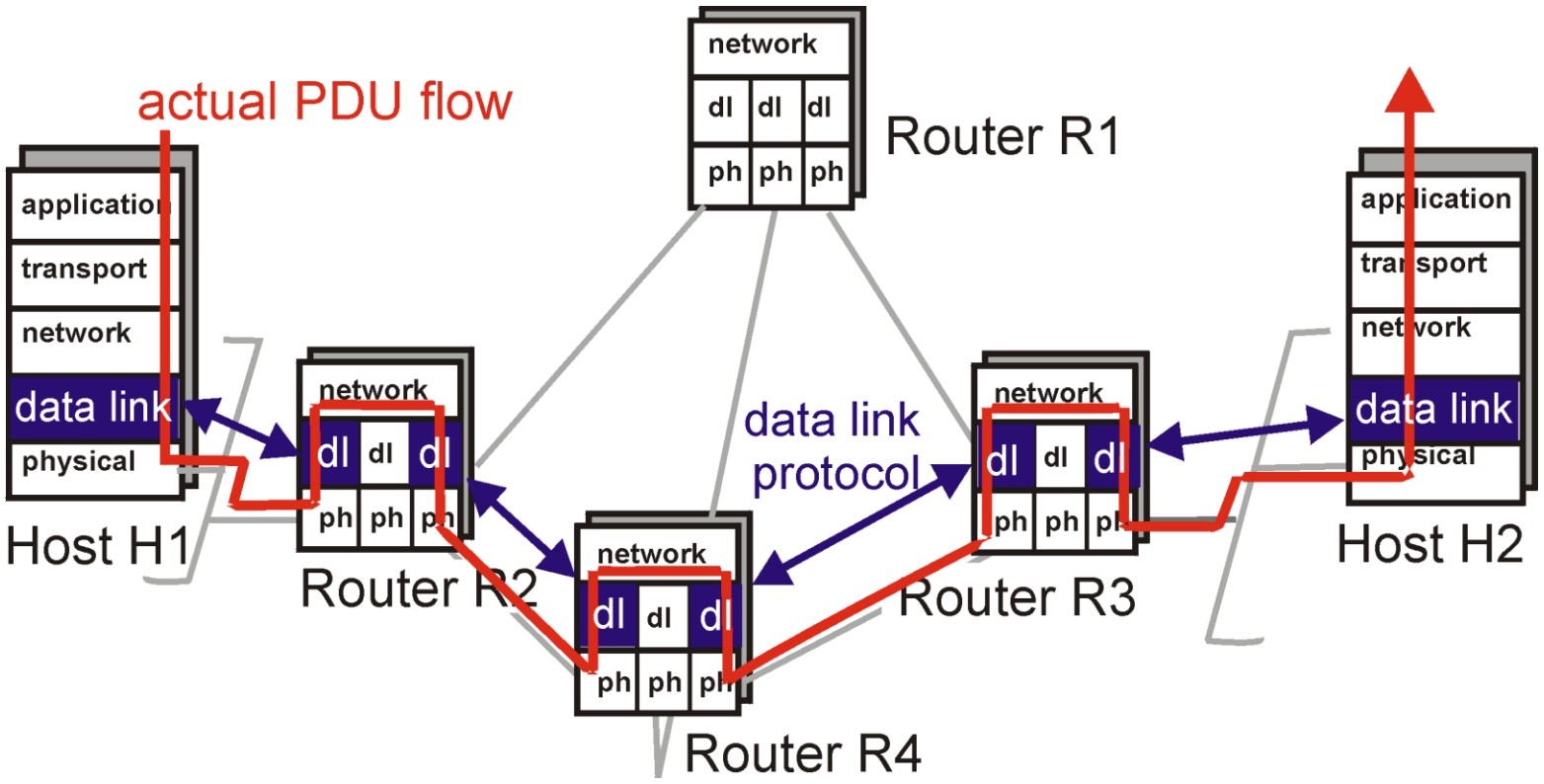
Link layer: context

- datagram transferred by different link protocols over different links:
  - e.g., Ethernet on first link, frame relay on intermediate links, 802.11 on last link
- each link protocol provides different services每个链路协议提供不同的服务
  - e.g., may or may not provide rdt over link

transportation analogy:

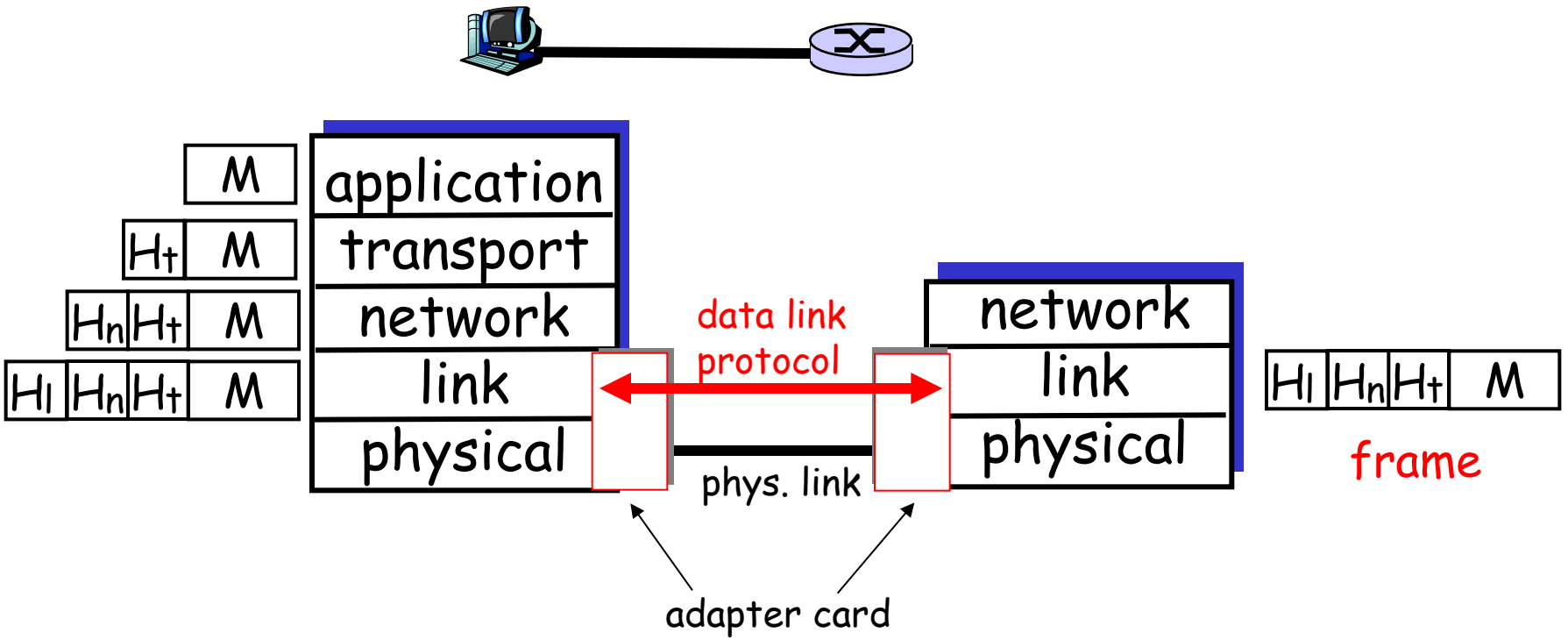
- trip from Princeton to Lausanne
  - car: Princeton to JFK
  - plane: JFK to Geneva
  - train: Geneva to Lausanne
- tourist = datagram
- transport segment = communication link
- transportation mode = link layer protocol
- travel agent = routing algorithm

Link Layer: setting the context



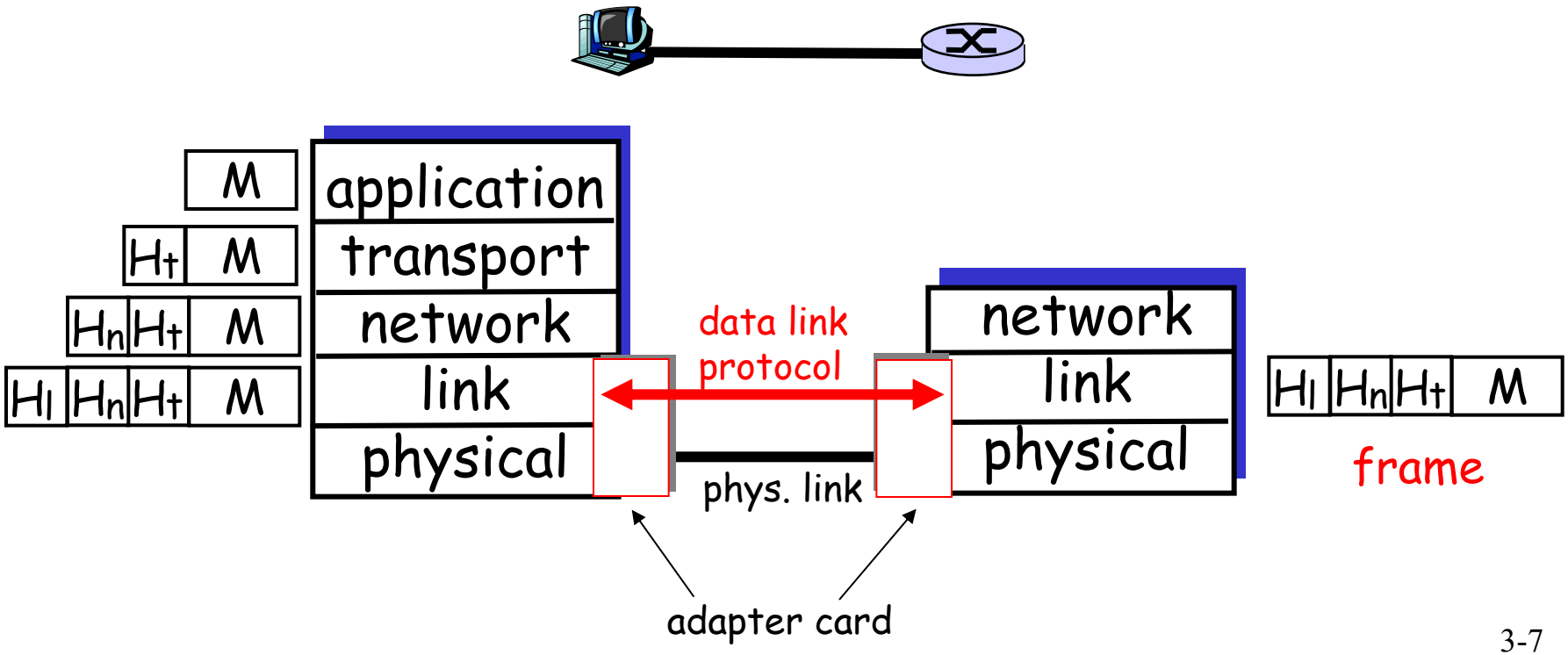
Link Layer: setting the context

- two *physically connected* devices:
  - host-router, router-router, host-host
- unit of data: *frame*



Link Layer: Implementation

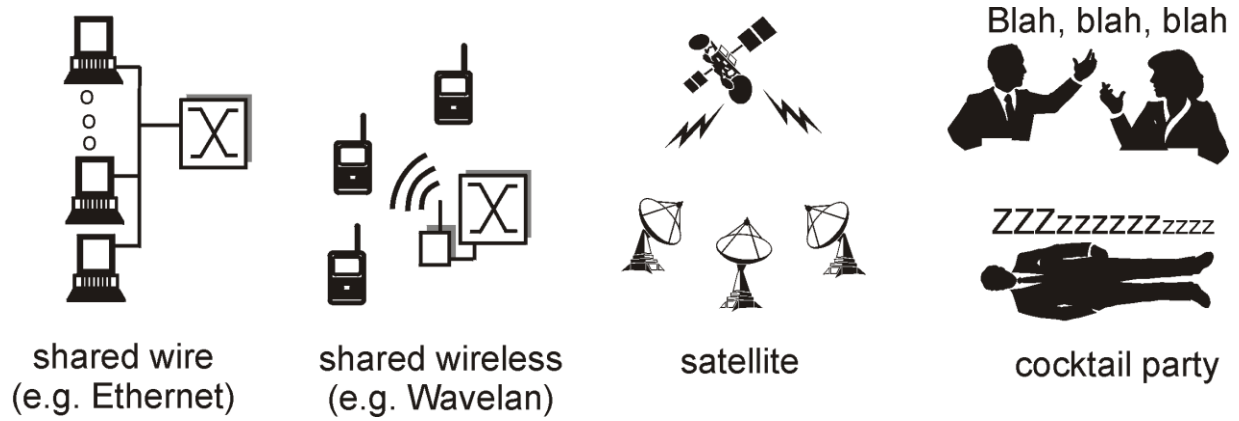
- implemented in "adapter"
  - e.g., PCMCIA card, Ethernet card
  - typically includes: RAM, DSP chips, host bus interface, and link interface



Multiple Access Links and Protocols多路访问链路和协议

Three types of "links":

- point-to-point (single wire, e.g. PPP, SLIP)
- broadcast (shared wire or medium; e.g, Ethernet, Wavelan, etc.)



- switched (e.g., switched Ethernet, ATM etc)

Link Layer Services

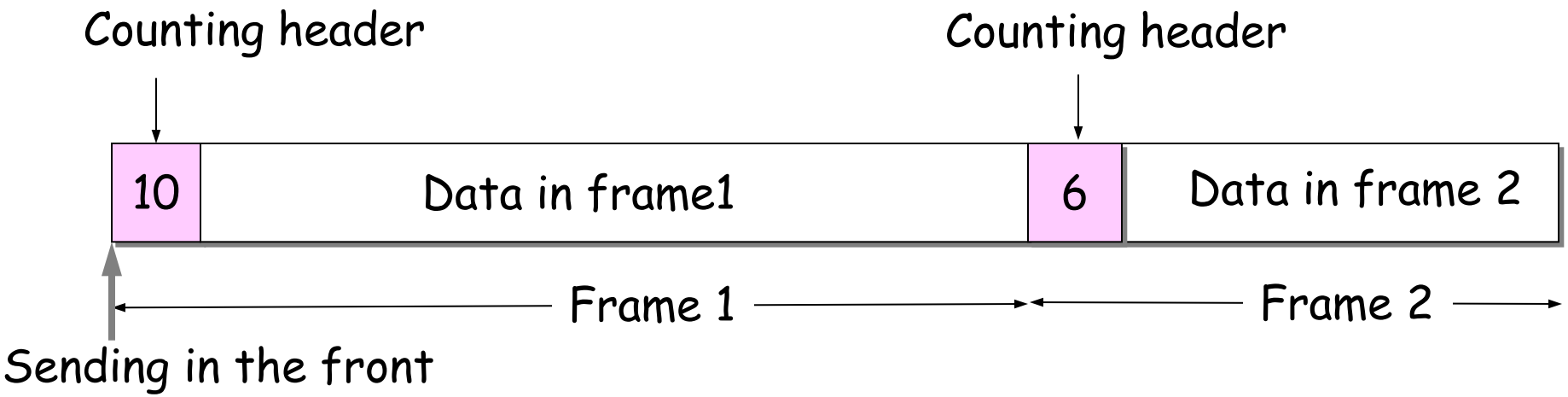
- Framing, link access:
  - encapsulate datagram into frame, adding header, trailer
  - 把数据表封装到帧中，添加帧头帧尾
  - implement channel access if shared medium, 共享介质的情况下
  - 'physical addresses' used in frame headers to identify source, dest 物理地址用于在帧头中识别源地址和目的地址
- Reliable delivery between two physically connected devices: (可靠交付)
  - Reliable data transfer protocol
  - seldom used on low bit error link (fiber, some twisted pair)在低比特差错链路中不常用
  - wireless links: high error rates常用于易发生高差错率的链路，例如无线链路

Link Layer Services (more)

- Flow Control: 流量控制
  - pacing between sender and receivers
- *Error Detection*: 差错检测
  - errors caused by signal attenuation, noise. 信号衰减和电磁噪声导致的错误
  - receiver detects presence of errors: 接收者检测到错误的存在
    - signals sender for retransmission or drops frame 重传或丢帧
- Error Correction: 差错纠正
  - receiver identifies *and corrects* bit error(s) without resorting to retransmission接收器识别并纠正比特错误，而无需重新传输

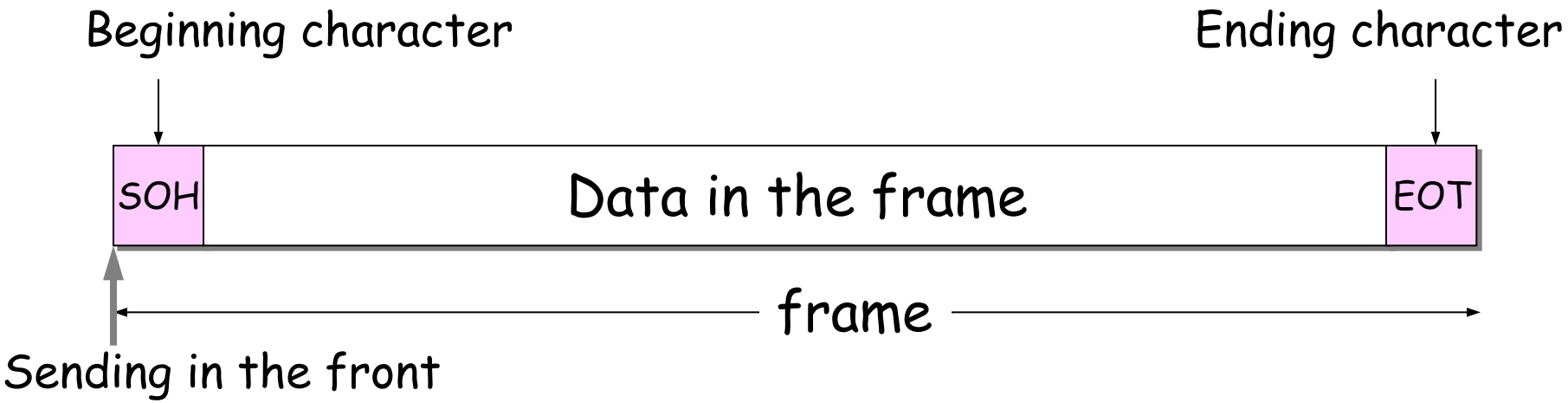
Framing

- Character count method : 字符计数法
  - encapsulate datagram into frame, adding header (character number)首部字段+数据字段



Framing

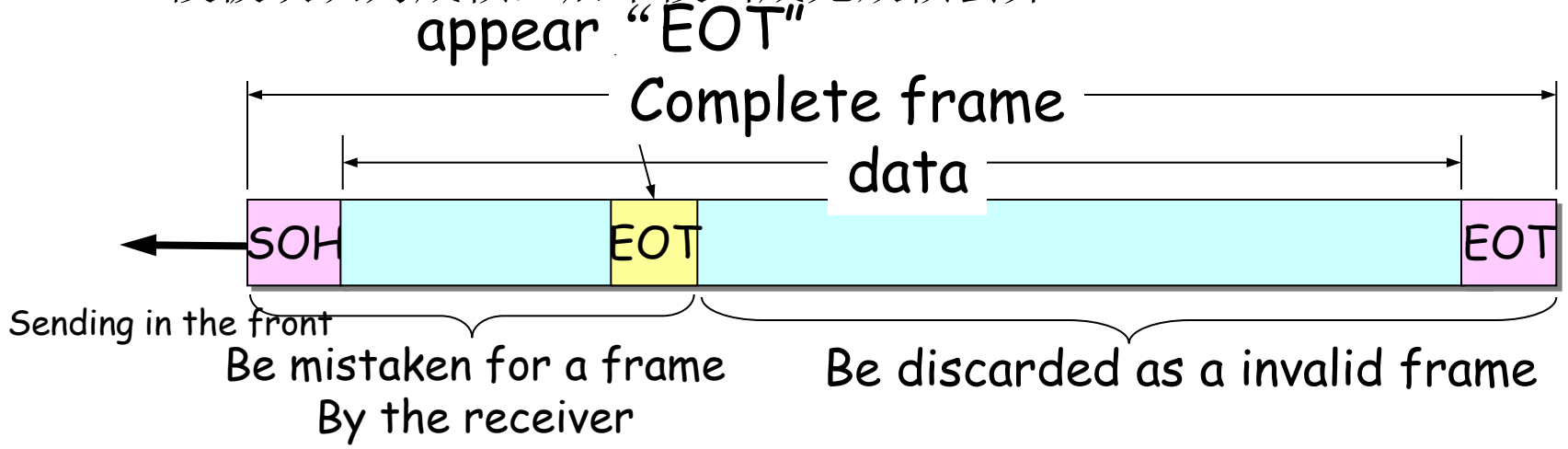
- First and tail bound method based on character:
  - encapsulate datagram into frame, adding header, trailer (character)首尾界定法



Framing

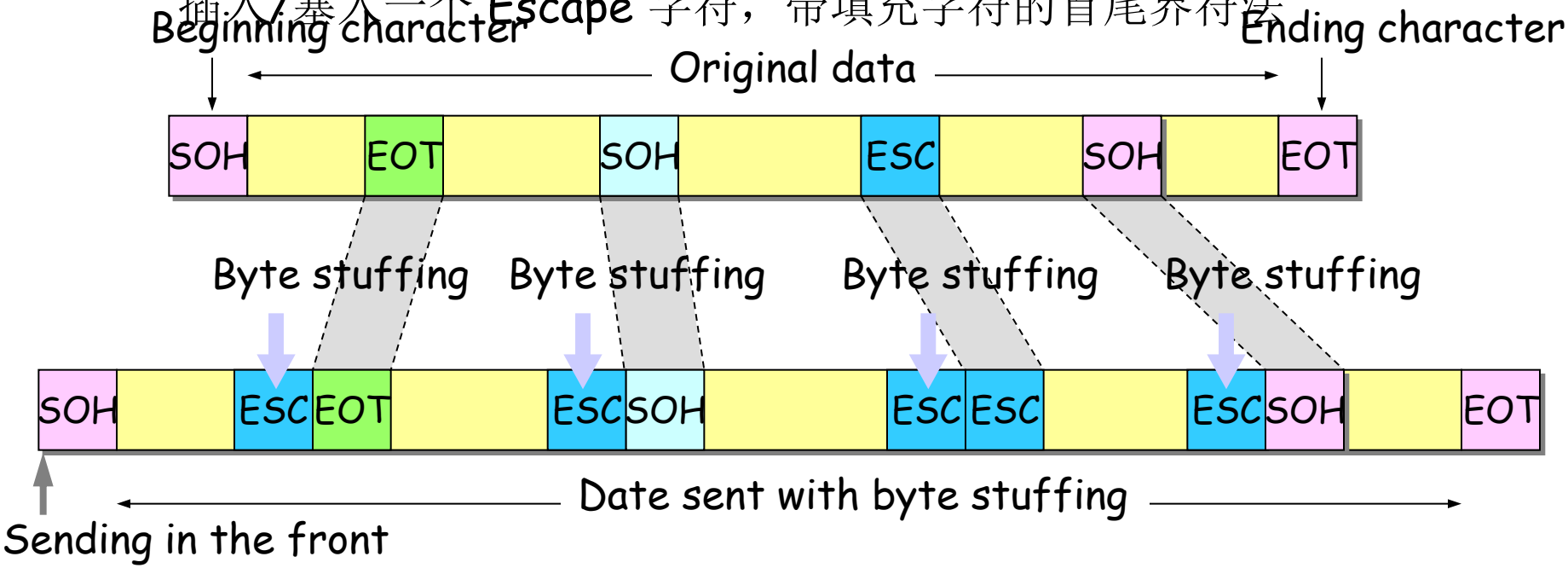
- First and tail bound method based on character:
  - encapsulate datagram into frame, adding header, trailer (character)

如果在传输的数据中出现和尾部字段相同的字符，可能会导致前半段被误认为成帧，后半段当成无效帧丢弃



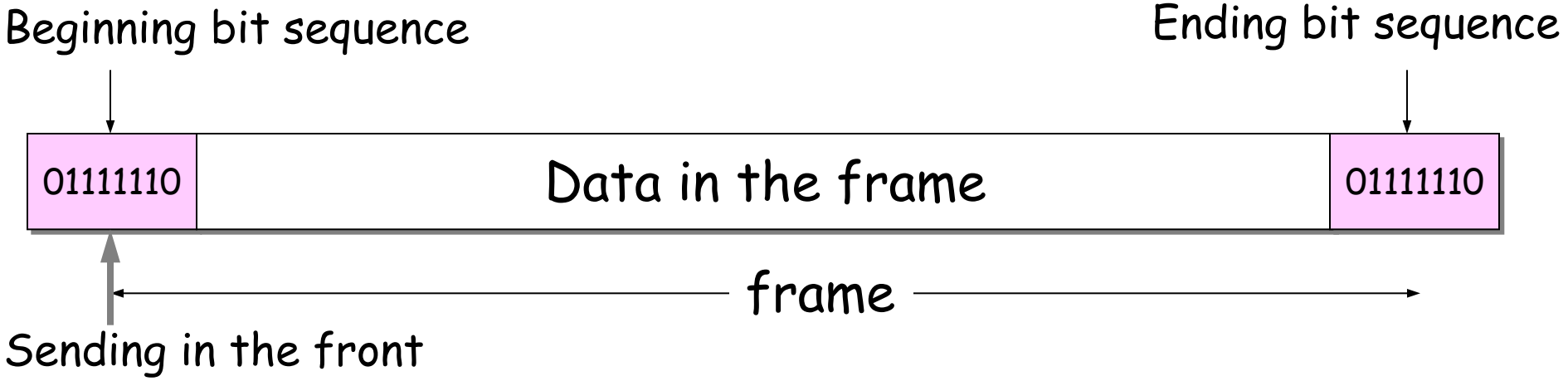
Framing

- First and tail bound method based on character:
  - Inserting/ stuffing a Escape character before the special character in the data part在数据部分的特殊字符前插入/塞入一个 Escape 字符，带填充字符的首尾界符法



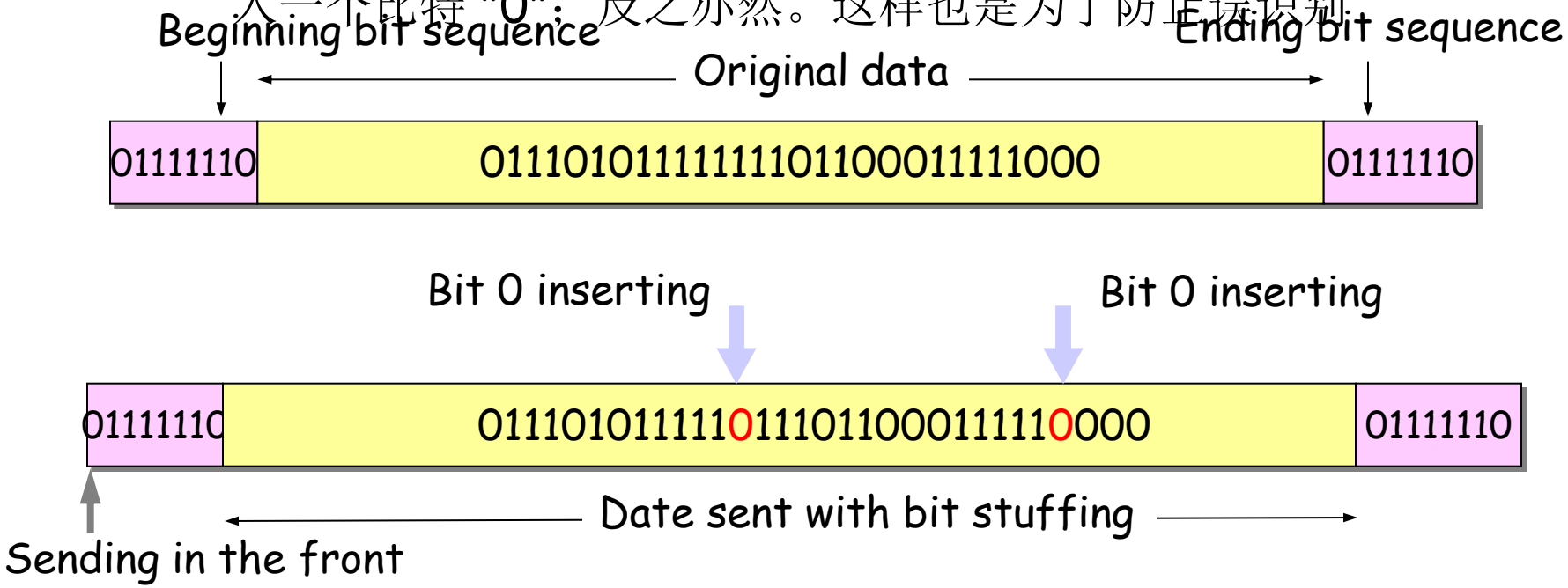
Framing

- First and tail bound method based on bit: 带位填充的首尾标记定界法encapsulate datagram into frame, adding header, trailer (bit sequence)起始和结束都用一个特殊的位串01111110



Framing

- First and tail bound method based on bit:
  - Inserting a bit "0" after successive five bits "0" in the transmitter; vice versa在发送器中连续五个比特 "0 "之后插入一个比特 "0"；反之亦然。这样也是为了防止误识别



Framing

- Physical layer coding violation method: 物理层编码违例法
  - encapsulate datagram into frame without stuffing无填充
  - Only be used in the networks with redundancy coding technology in the physical layer只适用于在物理媒体的编码策略中采用了冗余技术的网络
- For example (IEEE 802.11 with Manchester code):
  - Bit 1 with level jump mode from high to low
  - Bit 0 with level jump mode from low to high
  - Level jump modes from high to high or from low to low can be used for beginning and ending of a frame
  - 高-高/低-低不表示数据，因此可以用来做定界符

Exercise-1

1. HDLC protocol sets the bit stream 01111100001111110 after framing as (A?)

A. 011111000011111010

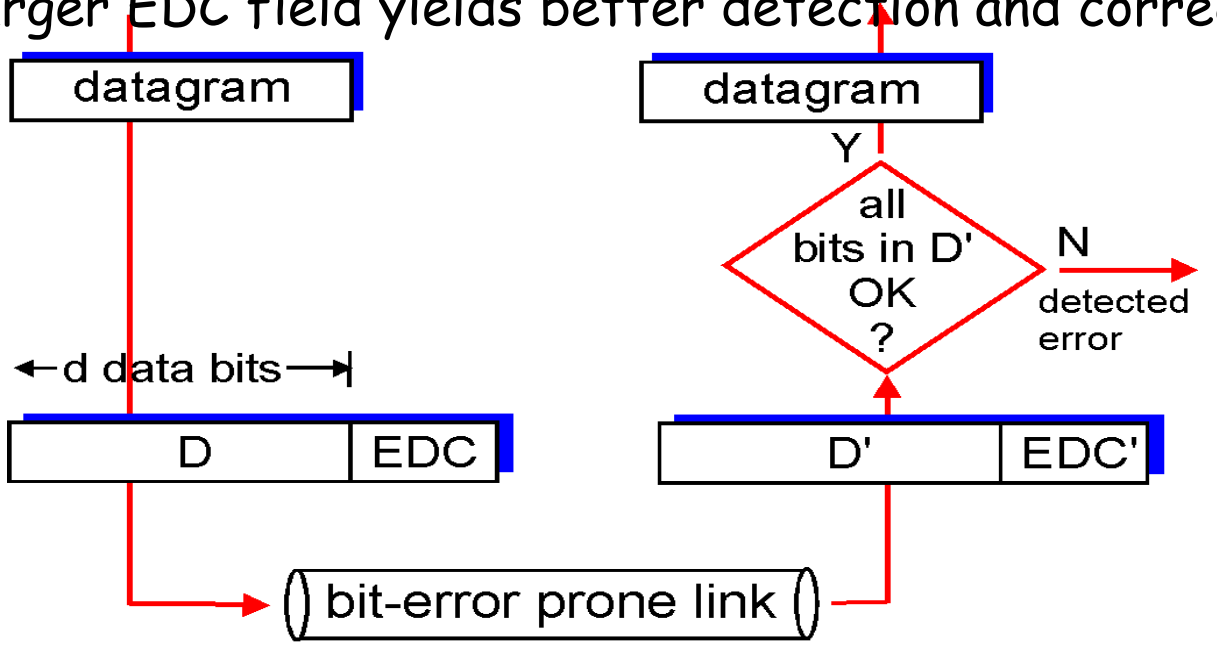
B.011111000111110101111110

C.011111000111111010

D.011111000111111001111101

Error Detection

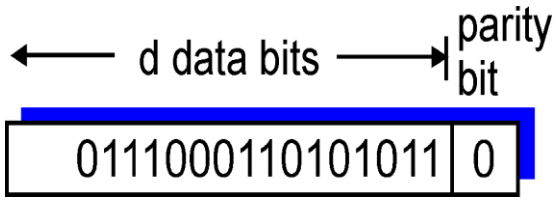
EDC= Error Detection and Correction bits (redundancy)
D = Data protected by error checking, may include header fields
D = 受错误检查保护的数据，可能包括标头字段
- Error detection not 100% reliable!
  - protocol may miss some errors, but rarely协议可能不是100%可靠
  - larger EDC field yields better detection and correction



Parity Checking

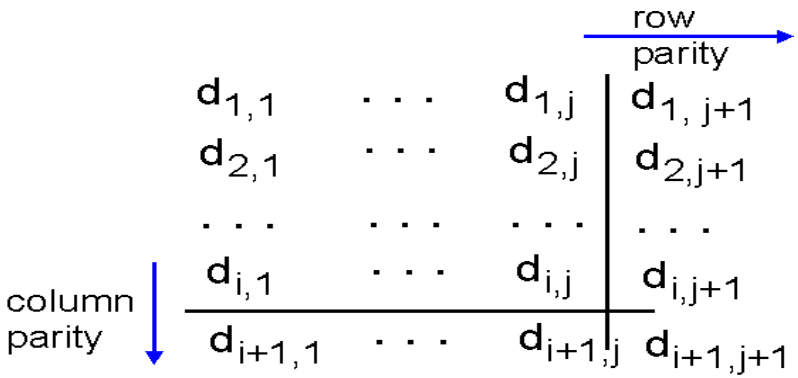
Single Bit Parity:

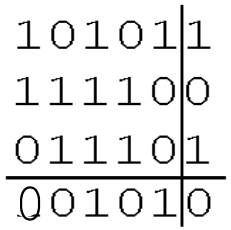
Detect single bit errors



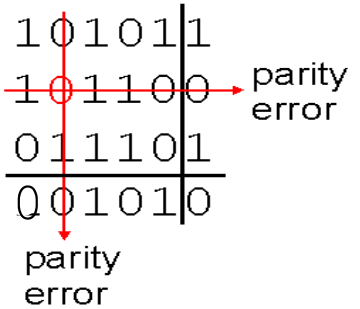
Two Dimensional Bit Parity:

Detect *and correct* single bit errors





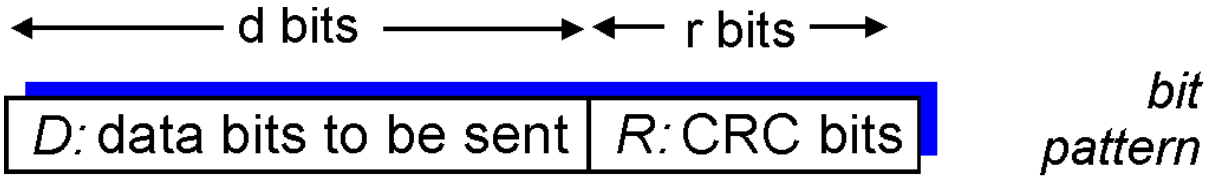
no errors



correctable single bit error

Checksumming: Cyclic Redundancy Check

- view data bits, D, as a binary number
- choose r+1 bit pattern (generator), G
- goal: choose r CRC bits, R, such that
  - <D,R> exactly divisible by G (modulo 2)
  - receiver knows G, divides <D,R> by G. If non-zero remainder: error detected!
  - can detect all burst errors less than r+1 bits
- widely used in practice (ATM, HDCL)



$D * 2^r$ XOR R

mathematical formula

CRC Example

Want:

D·2r XOR R = nG

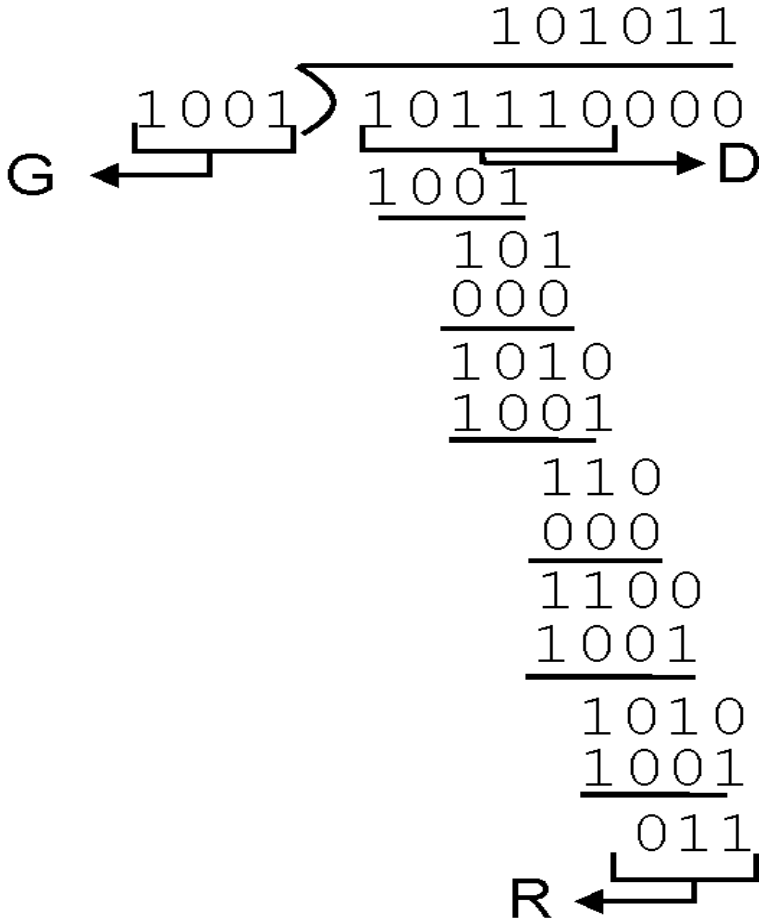
equivalently:

D·2r = nG XOR R

equivalently:

if we divide D·2r by G, want reminder R

$$R = \text{remainder}\left[\frac{D \cdot 2r}{G}\right]$$



CRC well-known generator

- CRC-12: $x^{12}+x^{11}+x^3+x^2+x^1+1$
- CRC-16 : $x^{16}+x^{15}+x^2+1$
- CRC-CCITT : $x^{16}+x^{12}+x^5+1$
- CRC-32: $x^{32}+x^{26}+x^{23}+x^{22}+x^{16}+x^{12}+x^{11}+x^{10}+x^8+x^7+x^5+x^4+x^2+x^1+1$

CCITT: Consultative Committee on International Telegraphy and Telephone

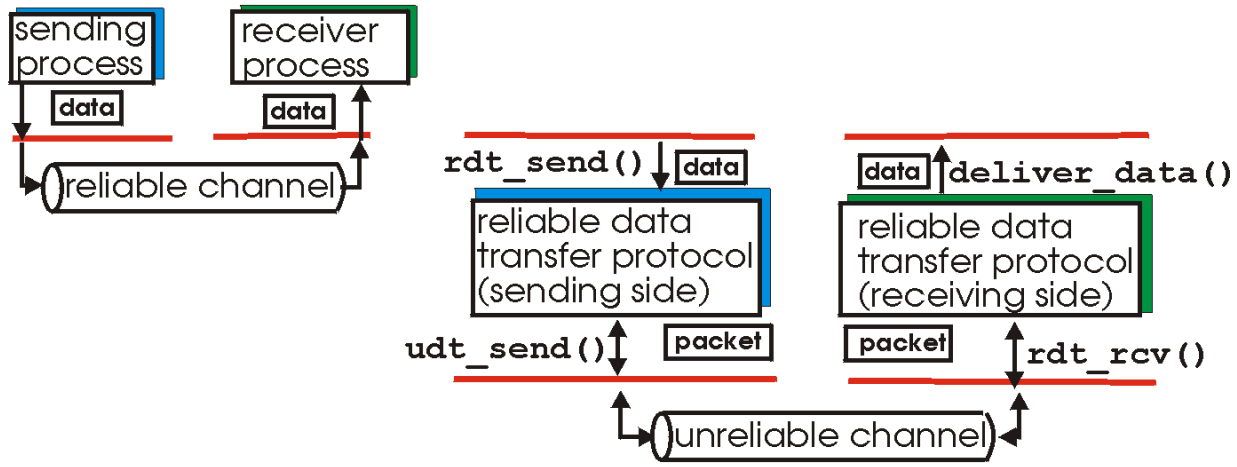
ITU-T: International Telecommunications Union – Telecommunications Standardization Sector

Exercise-2

1. If the information received by an Ethernet adapter is 101101001 and the generator is $G(x) = x^3+x^2+1$, judge whether the transmission has bit error?

Reliable Data Transfer可靠的数据传输

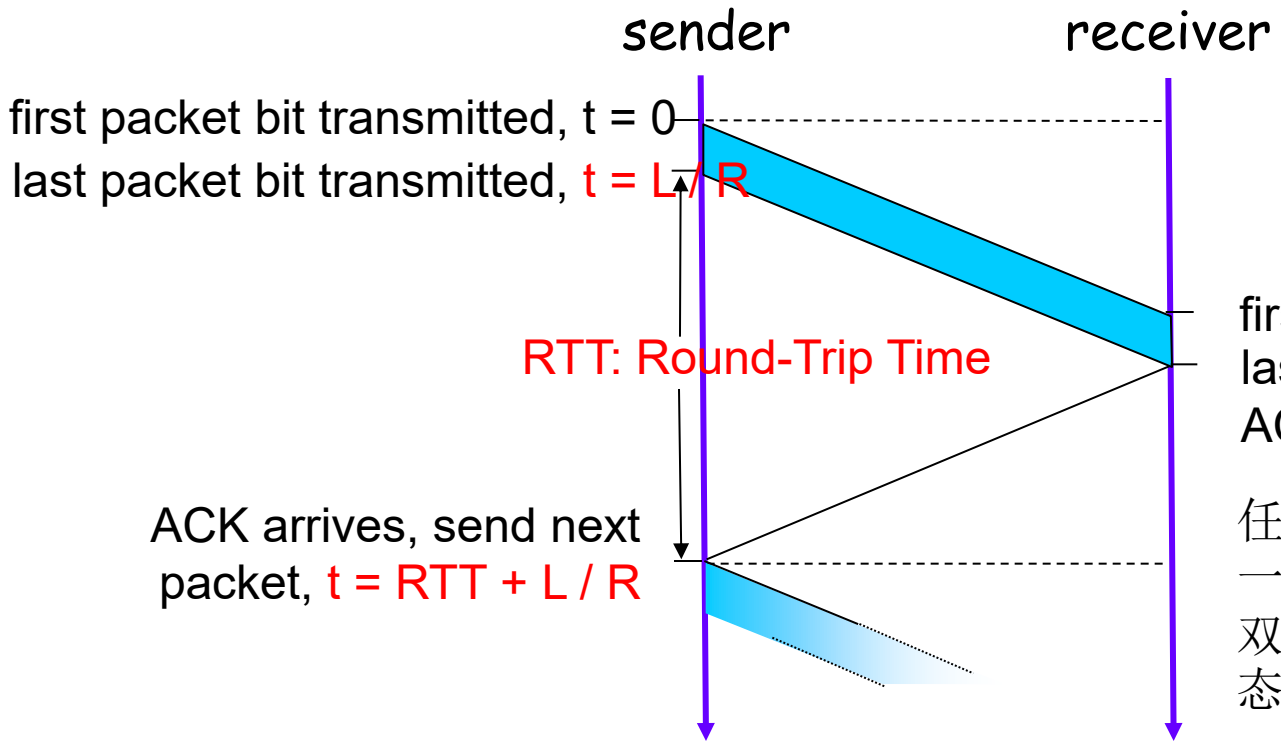
- Important in app., transport, link layers对应用层、传输层和链路层非常重要
- Top-10 list of important networking topics!



- Characteristics of unreliable channel will determine complexity of reliable data transfer protocol (rdt)不可靠信道的特性将决定可靠数据传输协议（RDT）的复杂性

Stop-and-Wait停止等待协议

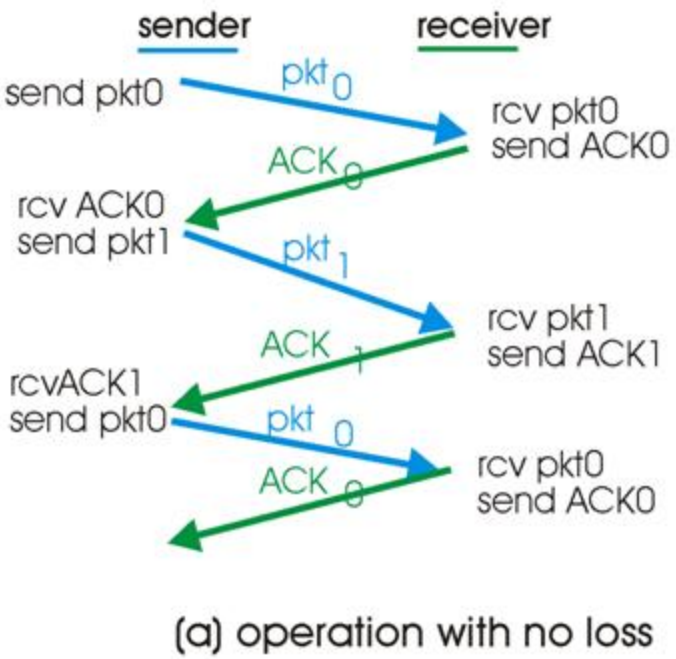
数据的流向是单向的，一方只发数据，另一方只发应答。发送方的数据称之为信息帧，接收方给出的数据称为应答帧，也叫*ACK*帧。



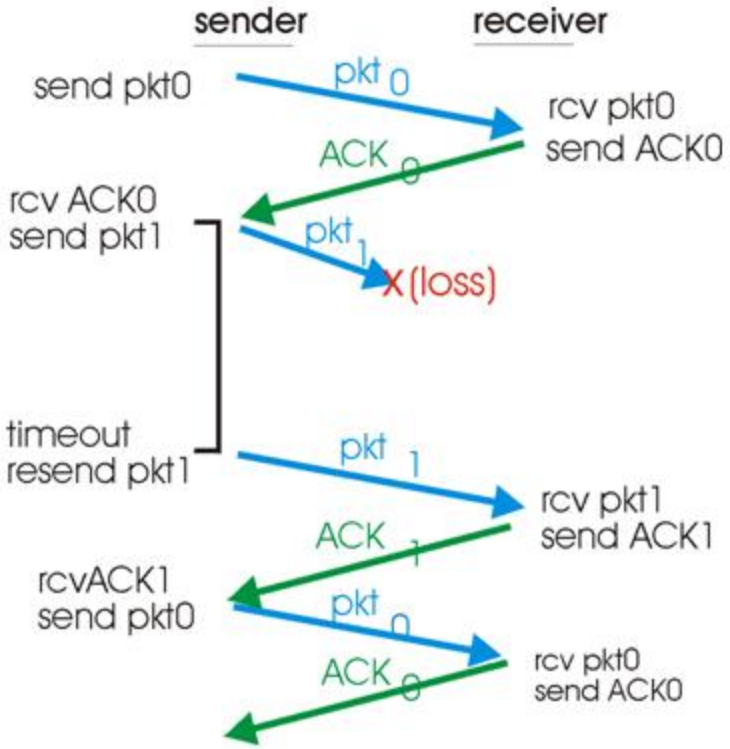
任意时刻，线路上都只有一种类型的帧传输，收发双方始终是一发一停的状态。

$$U_{sender} = \frac{L/R}{RTT + L/R} = \frac{.008}{30.008} = 0.00027$$

Stop-and-Wait

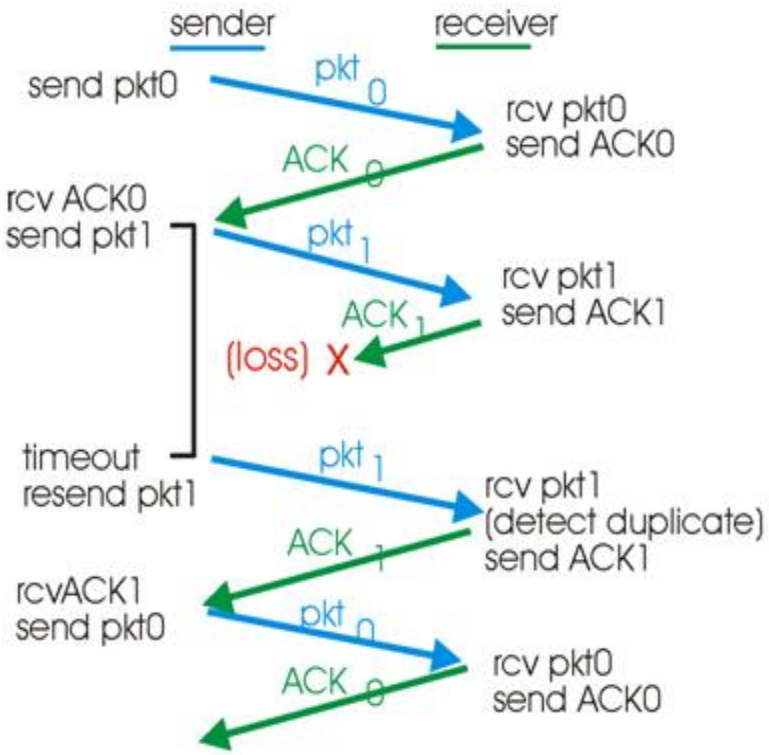


(a) operation with no loss

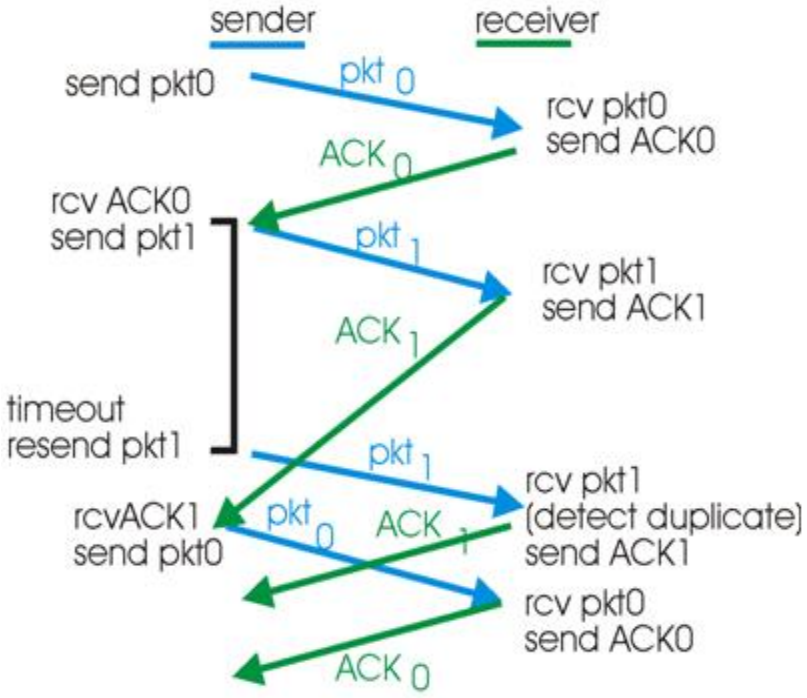


(b) lost packet

Stop-and-Wait



(c) lost ACK

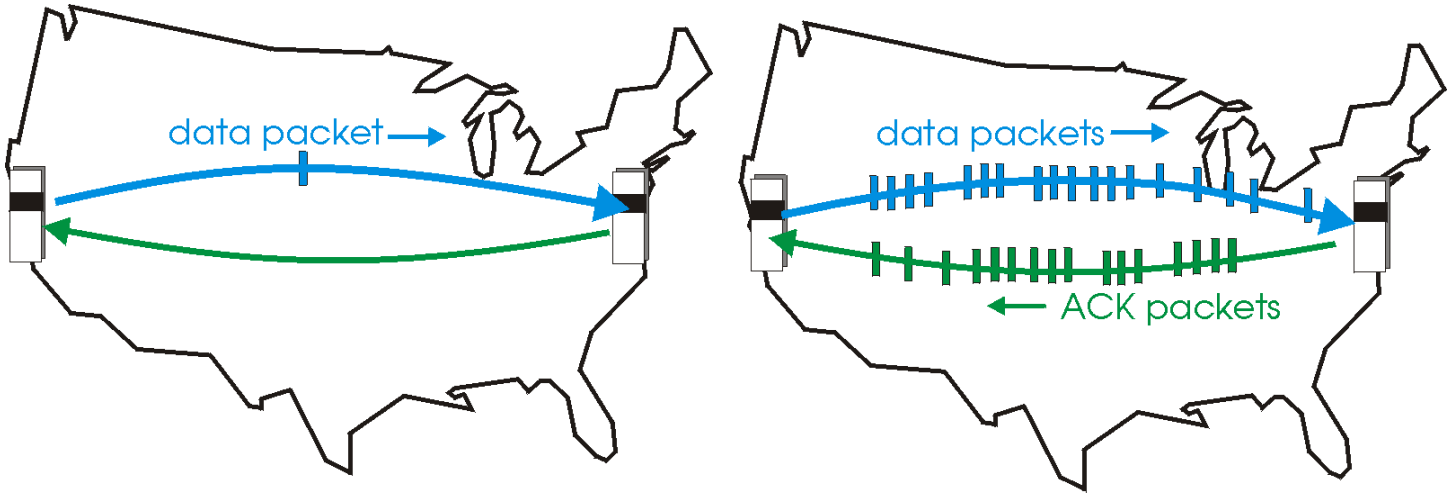


(d) premature timeout

Pipelined Protocols

Pipelining: sender allows multiple, "in-flight", yet-to-be-acknowledged pkts允许在上一个被确认之前就发送下一个

- Range of sequence numbers must be increased
- Buffering at sender and/or receiver



(a) a stop-and-wait protocol in operation

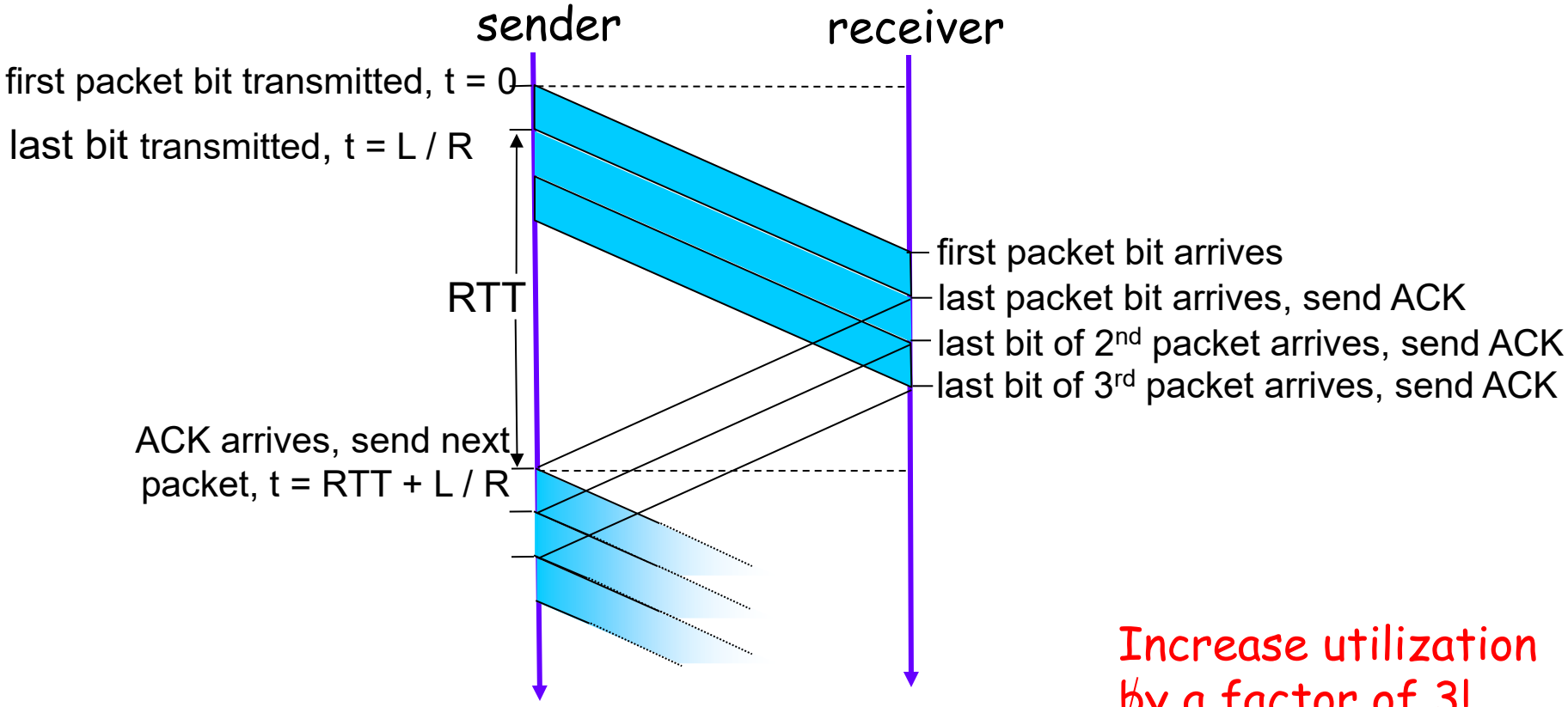
(b) a pipelined protocol in operation

Pipelined Protocols

- Advantage: much better bandwidth utilization than stop-and-wait带宽利用率远高于停等模式
  Disadvantage: More complicated to deal with reliability issues, e.g., corrupted, lost, out of order data.处理可靠性问题时更复杂，例如冲突、丢失等
  - Two generic approaches to solving this解决这一问题的两种通用方法
    - *Go-Back-N protocolsGo-Back-N协议(又称连续ARQ)*
    - *Selective repeat protocols选择重传ARQ协议*

- Note: *TCP is not exactly either*

Pipelining: Increased Utilization



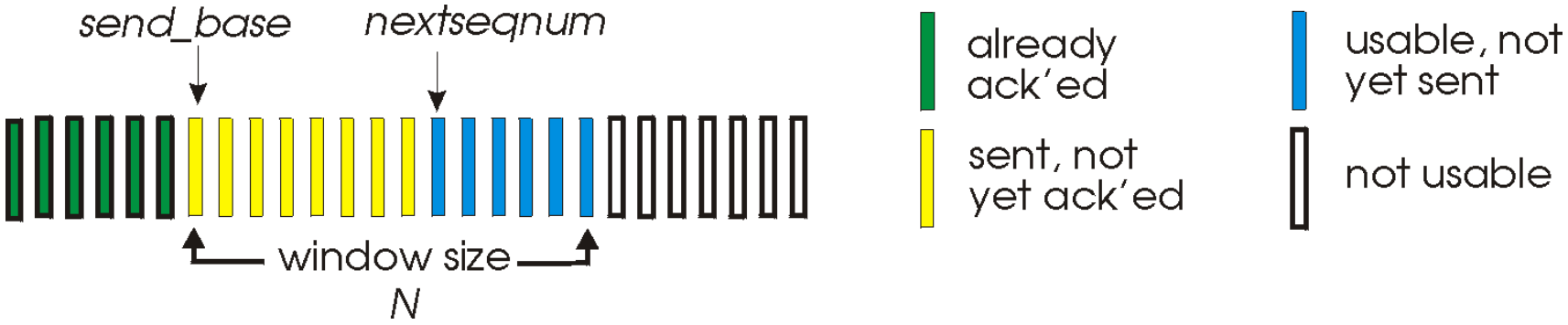
Increase utilization by a factor of 3!

$$U_{sender} = \frac{3 * L / R}{RTT + L / R} = \frac{.024}{30.008} = 0.0008$$

Go-Back-N

Sender:

- K-bit seq # in pkt header
- "Window" of up to N, consecutive unack'ed pkts allowed



- ACK(n): ACKs all pkts up to, including seq # n - "cumulative ACK"
  - May receive duplicate ACKs (see receiver)
- Timer for each in-flight pkt
- *Timeout(n)*: retransmit pkt n and all higher seq # pkts in window
- Called a sliding-window protocol滑动窗口

GBN: SenderGBN的发送方必须响应三种类型的事件

- rdt_Send() called: checks to see if window is full
  - No: send out packet
  - Yes: return data to upper level

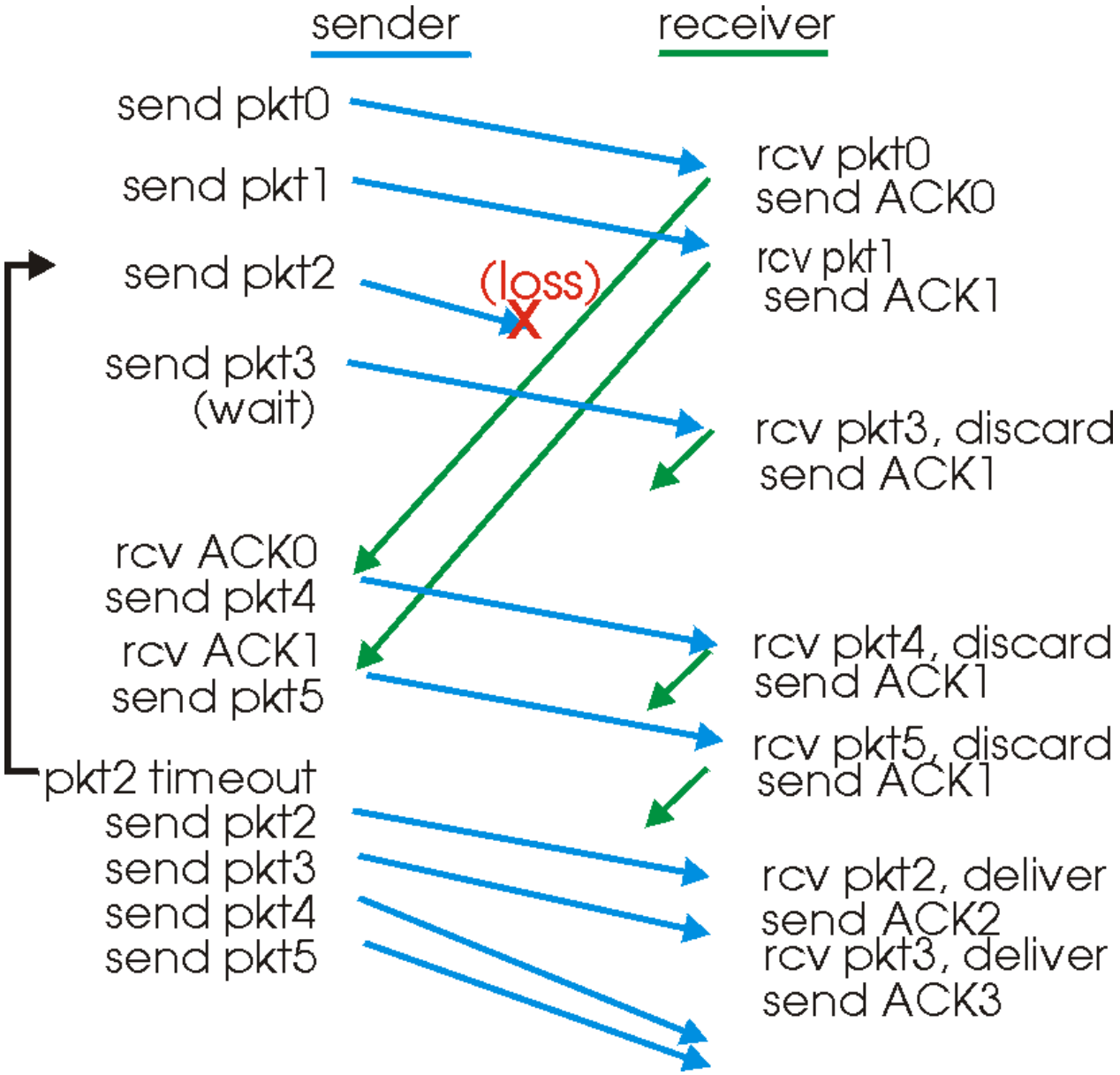
- Receipt of ACK(n): cumulative acknowledgement that all packets up to and including *n* have been received. Updates window accordingly and restarts timer

- Timeout: resends ALL packets that have been sent but not yet acknowledged.
  - This is only event that triggers resend

More on Receiver

- ACK-only: the receiver always sends ACK for last correctly received packet with highest *in-order* seq #
- Receiver only sends ACKS (no NAKs)
- Need only remember **`expectedseqnum`**
- Can generate duplicate ACKs

GBN In Action

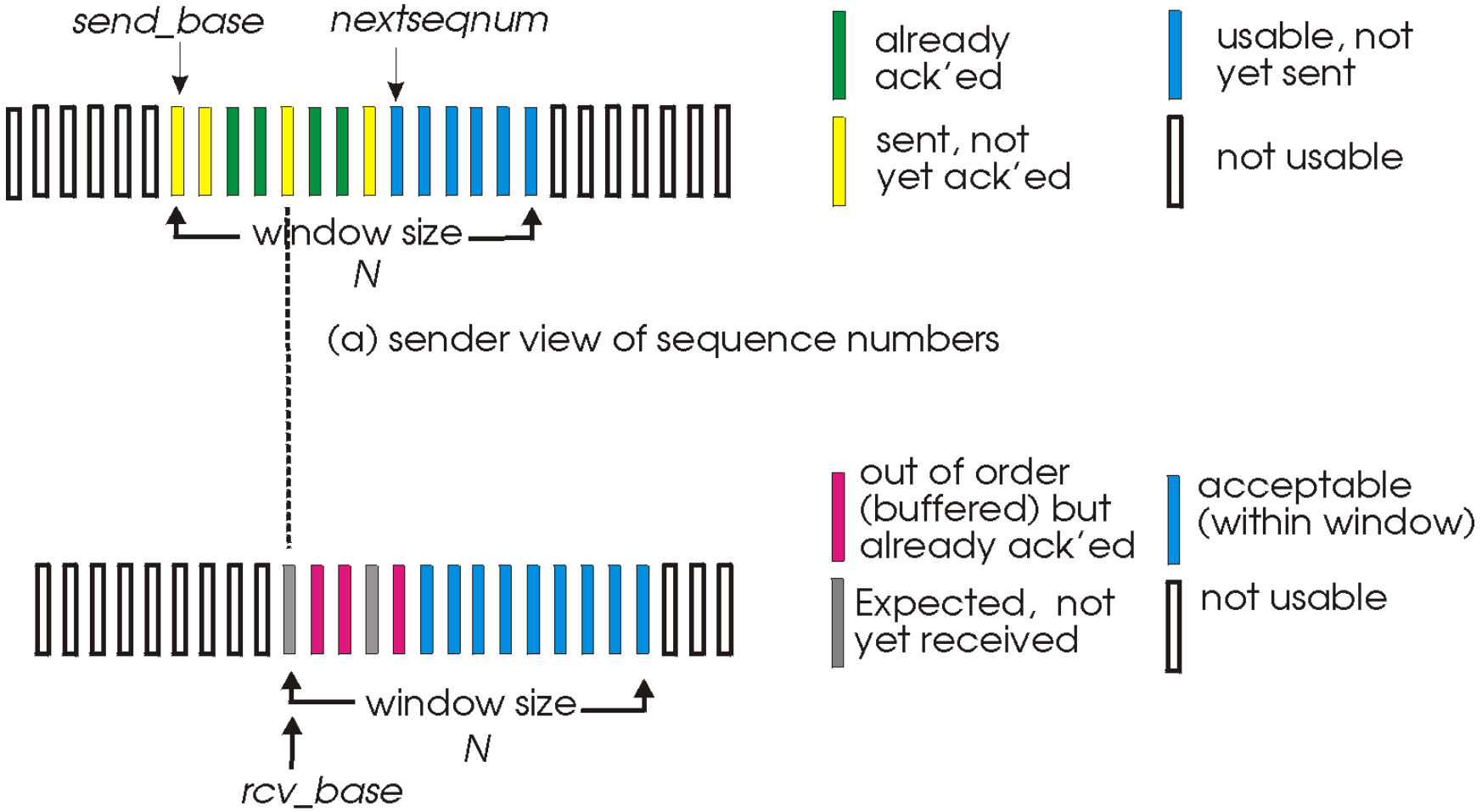


- GBN is easy to code but might have performance problems
- GBN 易于编码，但可能存在性能问题
- In particular, if many packets are in pipeline at one time (bandwidth-delay product large) then one error can force retransmission of huge amounts of data!
- 特别是，如果同时有许多数据包在管道中（带宽-延迟乘积较大），那么一个错误就会迫使重新传输大量数据！
- Selective Repeat protocol allows receiver to buffer data and only forces retransmission of required packets

Selective Repeat

- Receiver *individually* acknowledges all correctly received pkts
  - Buffers pkts, as needed, for eventual in-order delivery to upper layer
- Sender only resends pkts for which ACK not received
  - Sender timer for each unACKed pkt
  - Compare to GBN which only had timer for base packet
- Sender window
  - N consecutive seq #'s
  - Again limits seq #s of sent, unACKed pkts
  - Important: Window size < seq # range

Selective Repeat: Sender, Receiver Windows



Selective Repeat

sender

Data from above :

- If next available seq # in window, send pkt

Timeout(n):

- Resend pkt n, restart timer

ACK(n) in [sendbase,sendbase+N]:

- Mark pkt n as received
- If n smallest unACKed pkt, advance window base to next unACKed seq #

receiver

pkt n in [rcvbase, rcvbase+N-1]

- Send ACK(n)
- Out-of-order: buffer
- In-order: deliver (also deliver buffered, in-order pkts), advance window to next not-yet-received pkt

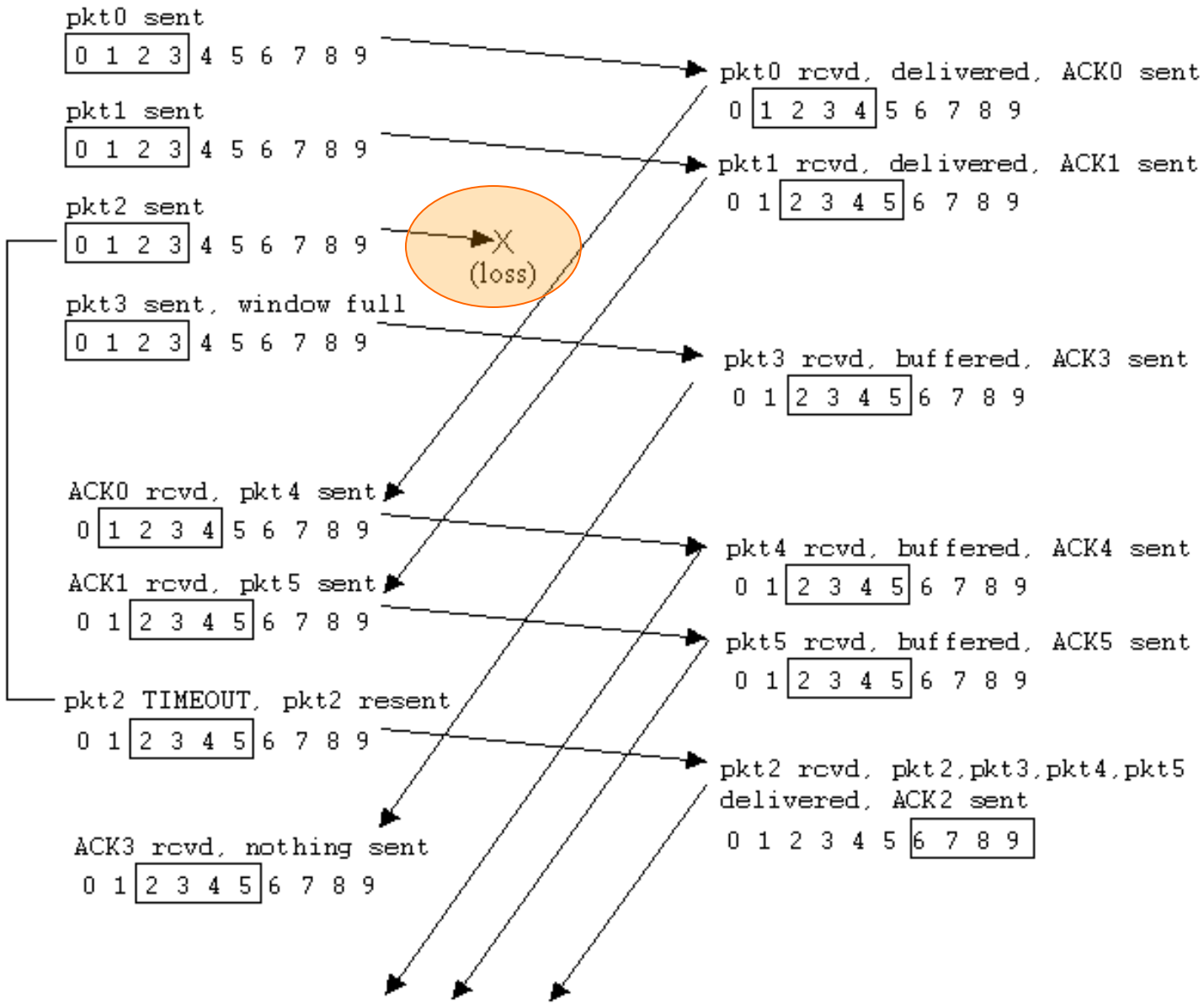
pkt n in [rcvbase-N,rcvbase-1]

- ACK(n)

Otherwise:

- Ignore

Selective Repeat In Action



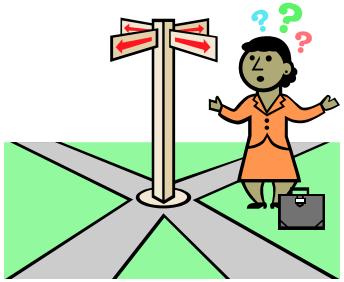
Selective Repeat: Dilemma

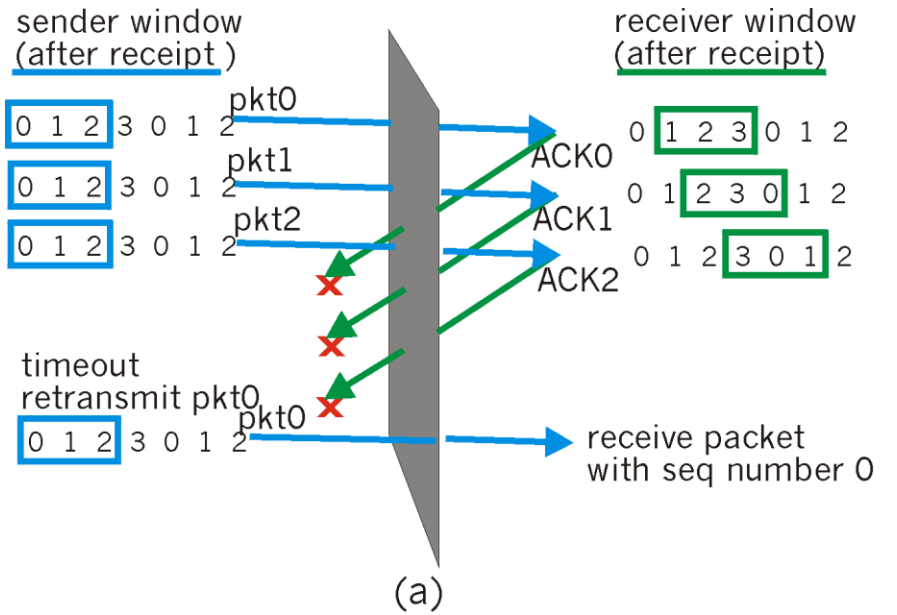
Example:

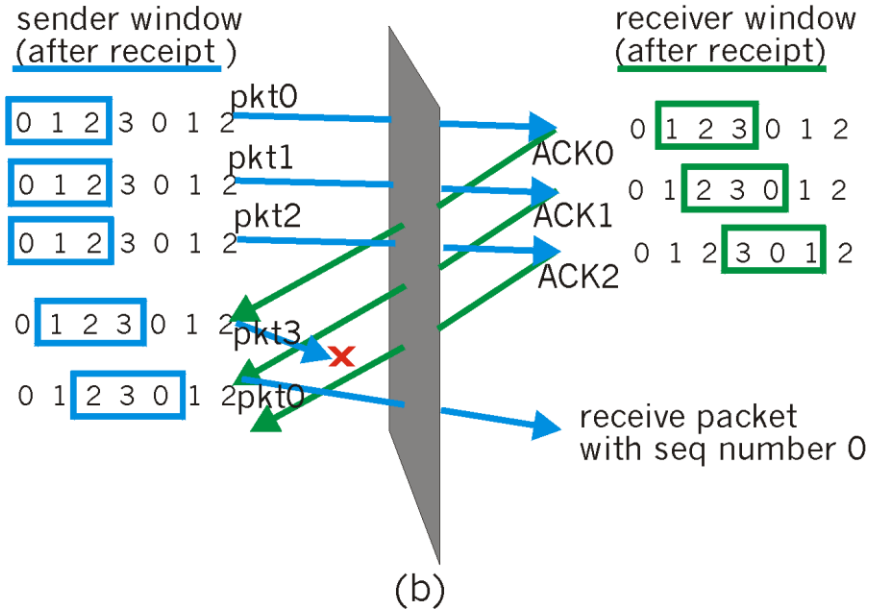
- seq #'s: 0, 1, 2, 3
- window size=3

- Receiver sees no difference in two scenarios!
- Incorrectly passes duplicate data as new in (a)

Q: what relationship between seq # size and window size?

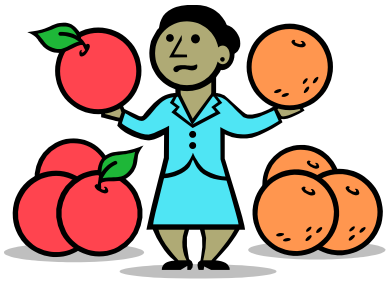






GBN vs. Selective Repeat

- Selective repeat is more complicated as it needs buffering at the receiver, but only retransmit packets required for retransmission



- GBN is simpler, but can lead to large number of unnecessary retransmission

Exercises-3

1. The data link layer uses the GBN protocol, and the sender has sent frames numbered 0 ~ 7. When the timer expires, if the sender only receives the acknowledgement of frames 0, 2 and 3, the number of frames to be retransmitted by the sender is ( )

A. 2 B. 3 C. 4 D.5

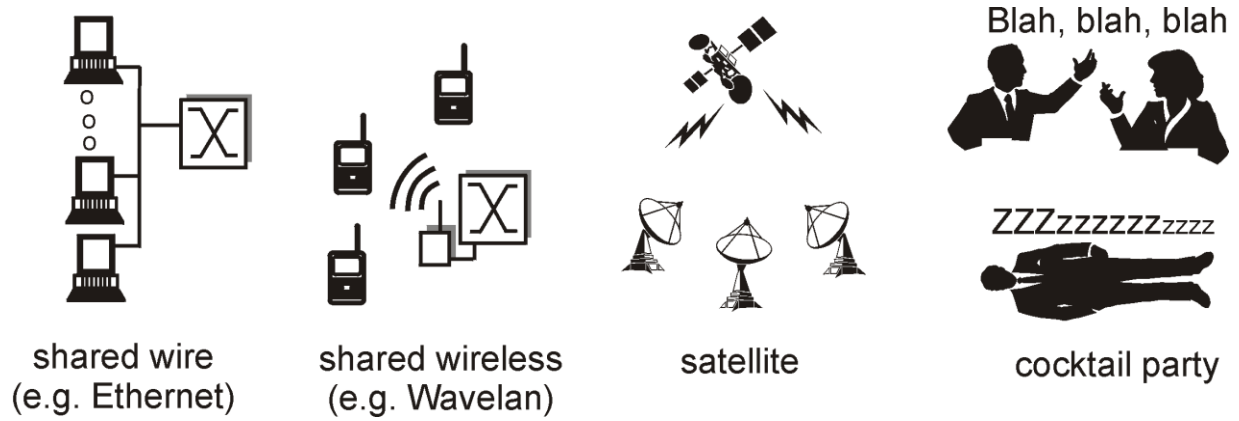
Exercise-3

2. Considering GBN protocol and SR protocol, assuming that the length of sequence number space is *n*, what is the maximum allowed sender window?

Multiple Access Links and Protocols

Three types of "links":

- point-to-point (single wire, e.g. PPP, SLIP)
- broadcast (shared wire or medium; e.g, Ethernet, Wavelan, etc.)



- switched (e.g., switched Ethernet, ATM etc)

Multiple Access protocols

- single shared communication channel
- two or more simultaneous transmissions by nodes: interference 节点同时进行两次或多次传输：干扰
- collision if node receives two or more signals at the same time
  - only one node can send successfully at a time 如果节点同时接收到两个或更多信号，则会发生碰撞，每次只有一个节点能成功发送信号
- *multiple access protocol:*
  - distributed algorithm that determines how nodes share channel, i.e., determine when node can transmit分布式算法，决定节点如何共享信道，即决定节点何时可以传输信号
  - communication about channel sharing must use channel itself!
  - what to look for in multiple access protocols:
    - synchronous or asynchronous 同步/异步
    - information needed about other stations 需要其他站点的信

Multiple Access protocols

- claim: humans use multiple access protocols all the time
- class can "guess" multiple access protocols
  - multiaccess protocol 1:
  - multiaccess protocol 2:
  - multiaccess protocol 3:
  - multiaccess protocol 4:

MAC Protocols: a taxonomy

Three broad classes:

- Channel Partitioning信道划分协议
  - divide channel into smaller "pieces" (time slots, frequency)
  - allocate piece to node for exclusive use
- Random Access随机接入协议
  - allow collisions
  - "recover" from collisions
- "Taking turns"轮流协议
  - tightly coordinate shared access to avoid collisions

Goal: efficient, fair, simple, decentralized

Channel Partitioning MAC protocols: TDMA

TDMA: time division multiple access 时分多路复用

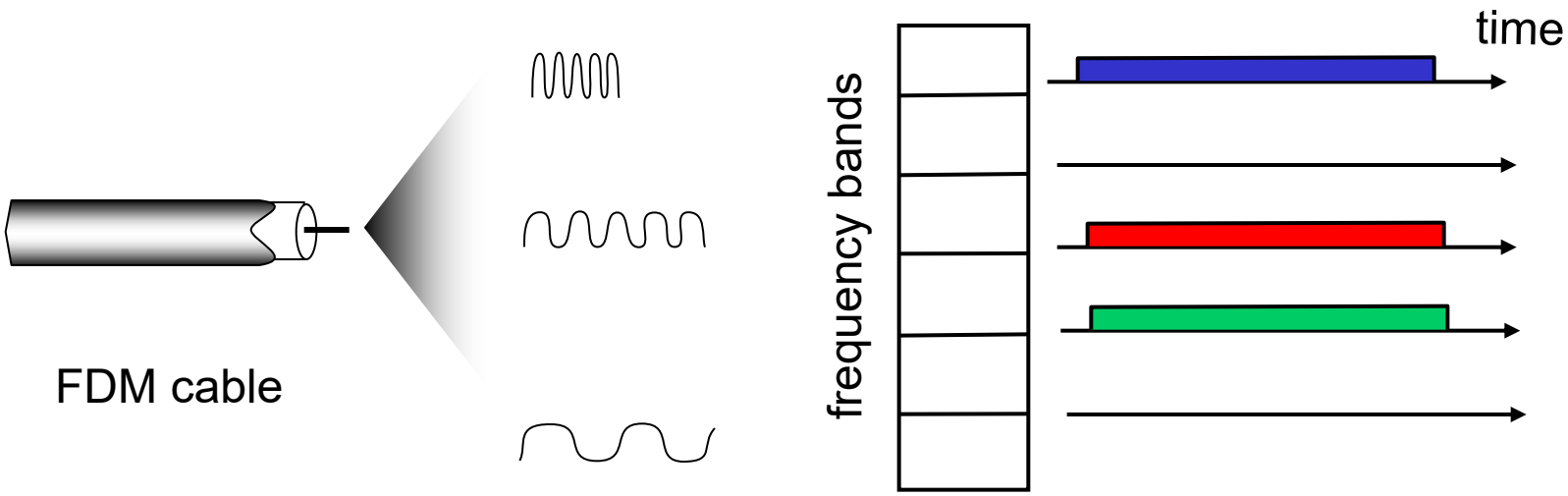
- access to channel in "rounds" （周期性地，一轮一轮来）
- each station gets fixed length slot (length = pkt trans time) in each round 一个时隙内应允许传输单个分组。
- unused slots go idle 没用的时隙闲置
- example: 6-station LAN, 1,3,4 have pkt, slots 2,5,6 idle



Channel Partitioning MAC protocols: FDMA

FDMA: frequency division multiple access

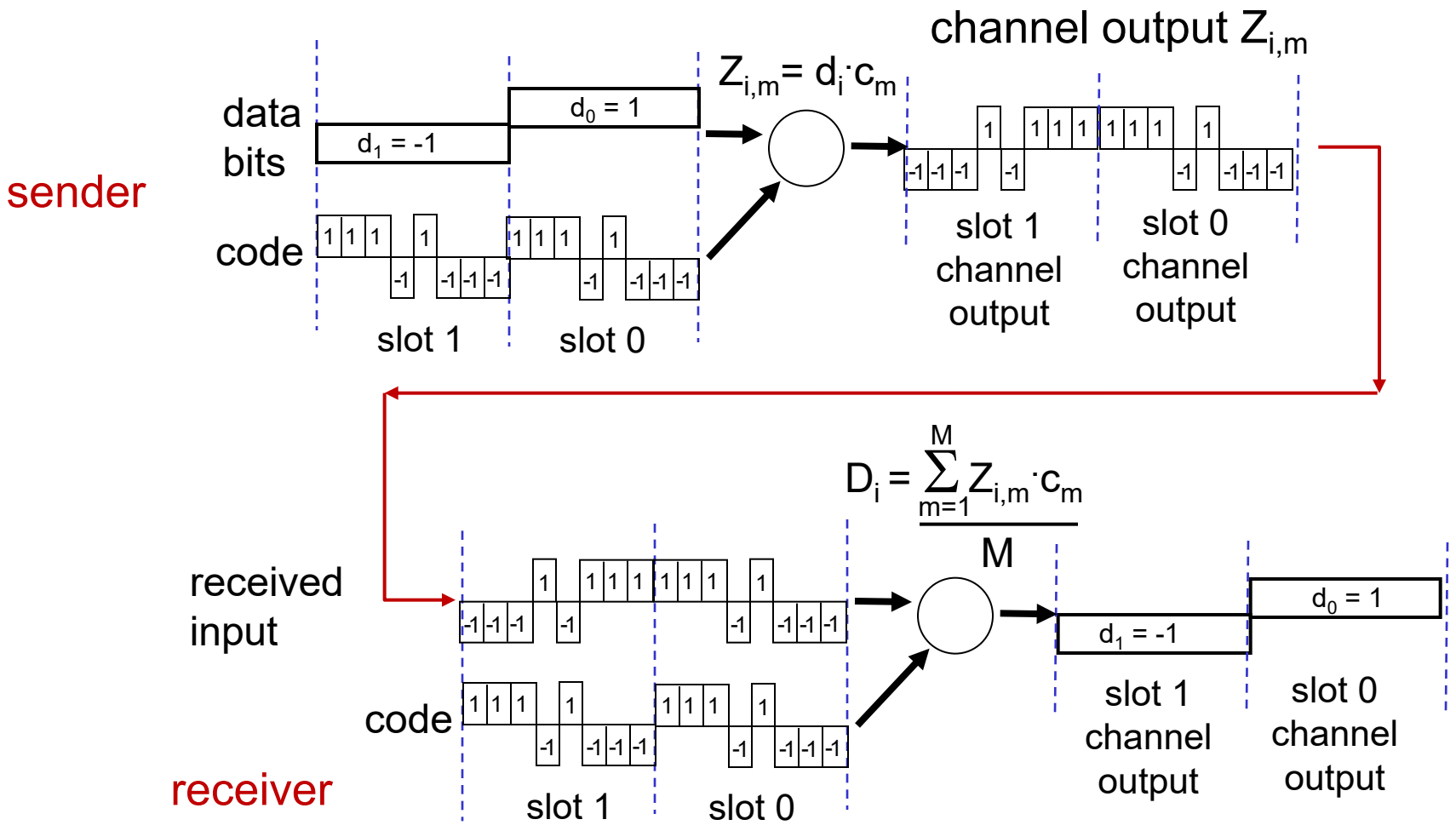
- channel spectrum divided into frequency bands
- each station assigned fixed frequency band
- unused transmission time in frequency bands go idle
- example: 6-station LAN, 1,3,4 have pkt, frequency bands 2,5,6 idle



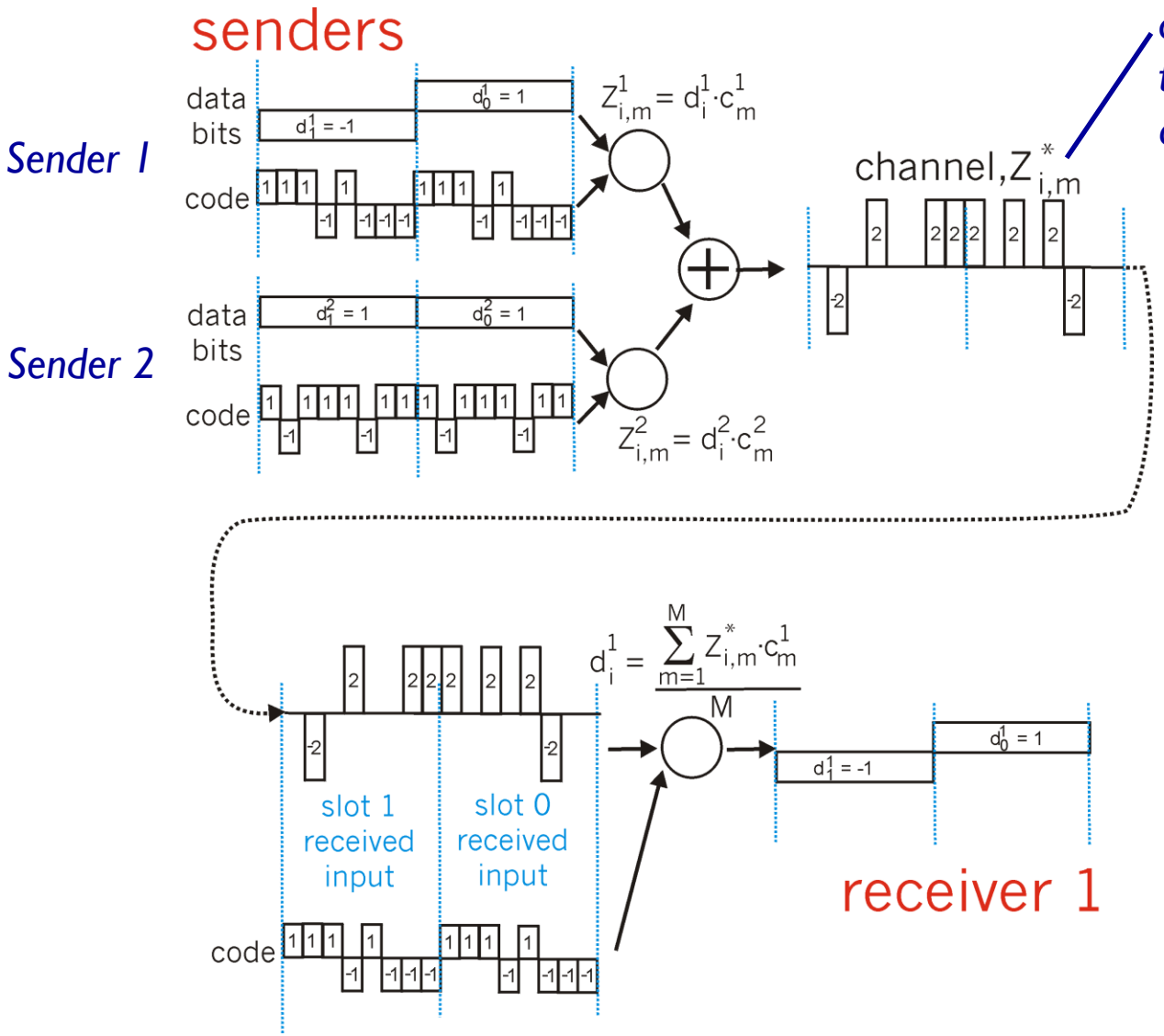
Code Division Multiple Access (CDMA)

- unique "code" assigned to each user; i.e., code set partitioning对每一个节点分配一种不同的编码
  - all users share same frequency, but each user has own "chipping" sequence (i.e., code) to encode data所有用户共享相同的频率，但每个用户都有自己的 "芯片 "序列（即代码）对数据进行编码
  - allows multiple users to "coexist" and transmit simultaneously with minimal interference (if codes are "orthogonal")
- *encoded signal* = (original data) X (chipping sequence)
- *decoding:* inner-product of encoded signal and chipping sequence

CDMA encode/decode



CDMA: two-sender interference



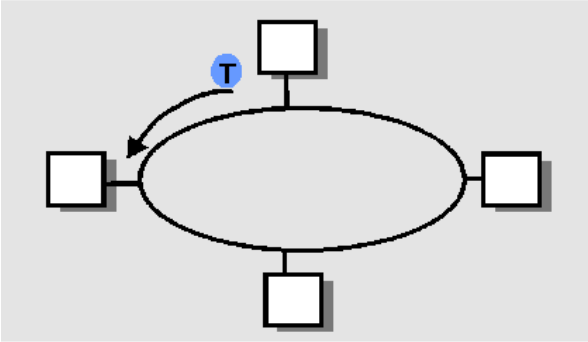
"Taking Turns" MAC protocols

Polling: 投票

- master node "invites" slave nodes to transmit in turn
- Request to Send, Clear to Send msgs
- concerns:
  - polling overhead
  - latency
  - single point of failure (master)

Token passing:

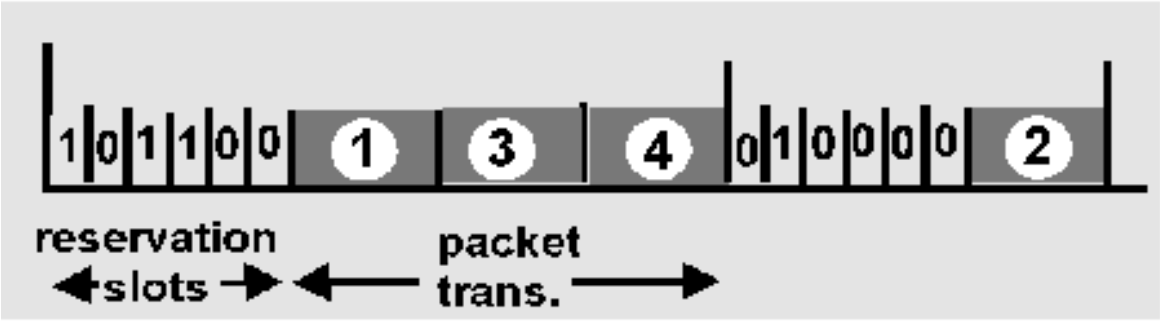
- control **token** passed from one node to next sequentially.
- token message
- concerns:
  - token overhead
  - latency
  - single point of failure (token)



Reservation-based protocols

Distributed Polling:

- time divided into slots
- begins with N short reservation slots
  - reservation slot time equal to channel end-end propagation delay
  - station with message to send posts reservation
  - reservation seen by all stations
- after reservation slots, message transmissions ordered by known priority

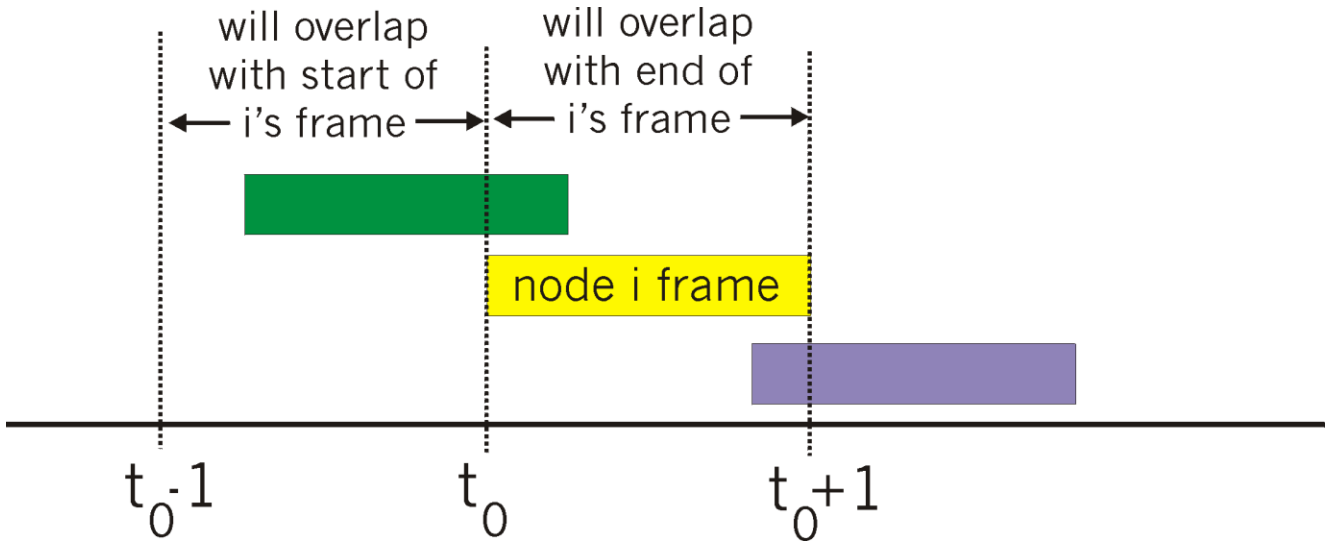


Random Access protocols

- When node has packet to send
  - transmit at full channel data rate R.
  - no *a priori* coordination among nodes
- two or more trasnmitting nodes -> "collision",
- random access MAC protocol specifies:
  - how to detect collisions
  - how to recover from collisions (e.g., via delayed retransmissions)
- Examples of random access MAC protocols:
  - ALOHA
  - slotted ALOHA
  - CSMA and CSMA/CD

Pure ALOHA

- unslotted Aloha: simpler, no synchronization
- pkt needs transmission:
  - send without awaiting for beginning of slot
- collision probability increases:
  - pkt sent at t_0 collide with other pkts sent in $[t_0-1, t_0+1]$



Pure Aloha (cont.)

P(success by given node) = P(node transmits) ·

P(no other node transmits in $[p_0-1,p_0]$ ·

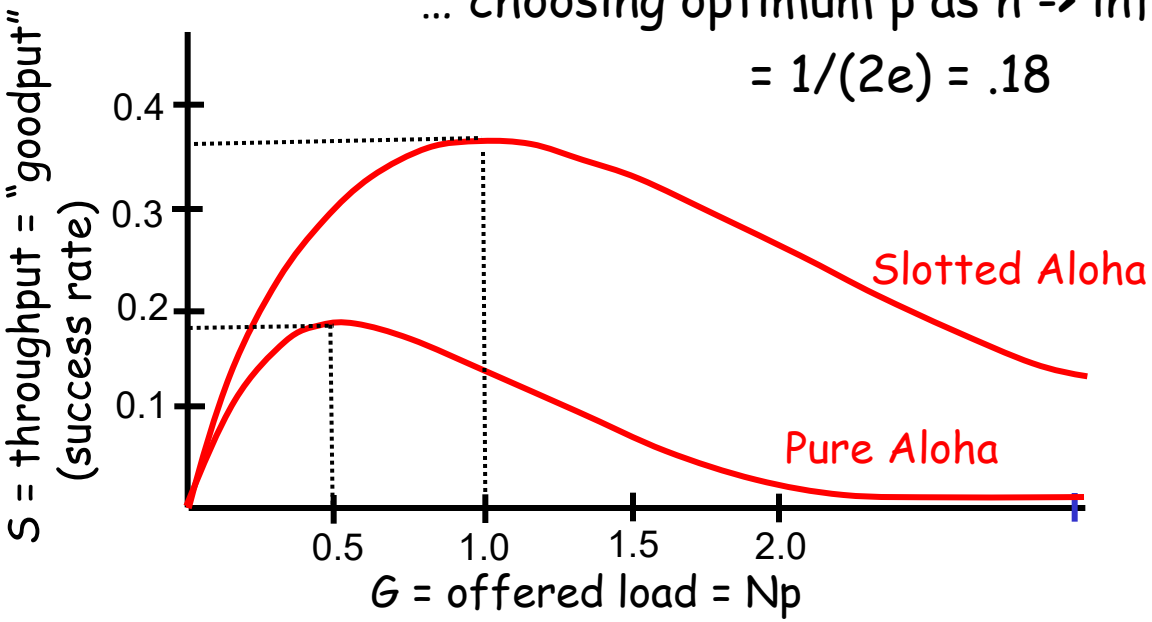
P(no other node transmits in $[p_0-1,p_0]$

$= p \cdot (1-p)^{N-1} \cdot (1-p)^{N-1}$

$= p \cdot (1-p)^{2(N-1)}$

… choosing optimum p as n -> infty …

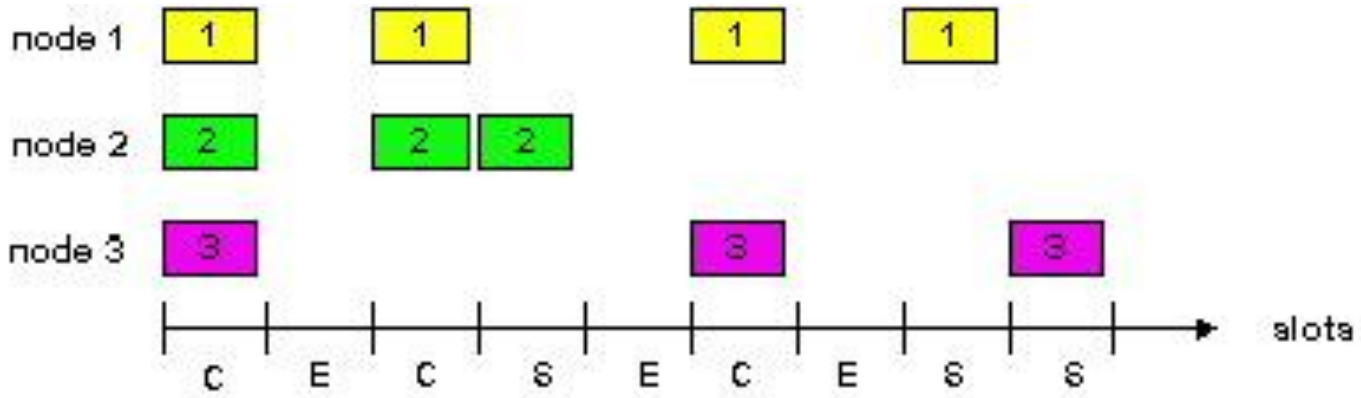
$= 1/(2e) = .18$



protocol constrains effective channel throughput!

Slotted Aloha

- time is divided into equal size slots (= pkt trans. time)
- node with new arriving pkt: transmit at beginning of next slot
- if collision: retransmit pkt in future slots with probability p, until successful.



Success (S), Collision (C), Empty (E) slots

Slotted Aloha efficiency

Q: what is max fraction slots successful?

A: Suppose N stations have packets to send

- each transmits in slot with probability *p*
- prob. successful transmission *S* is:

by single node: $S = p\,(1-p)^{(N-1)}$

by any of N nodes

S = Prob (only one transmits)

$= N\,p\,(1-p)^{(N-1)}$

... choosing optimum p as n -> infty ...

= 1/e = .37 as N -> infty

At best: channel use for useful transmissions 37% of time!

CSMA: Carrier Sense Multiple Access)

CSMA: listen before transmit:

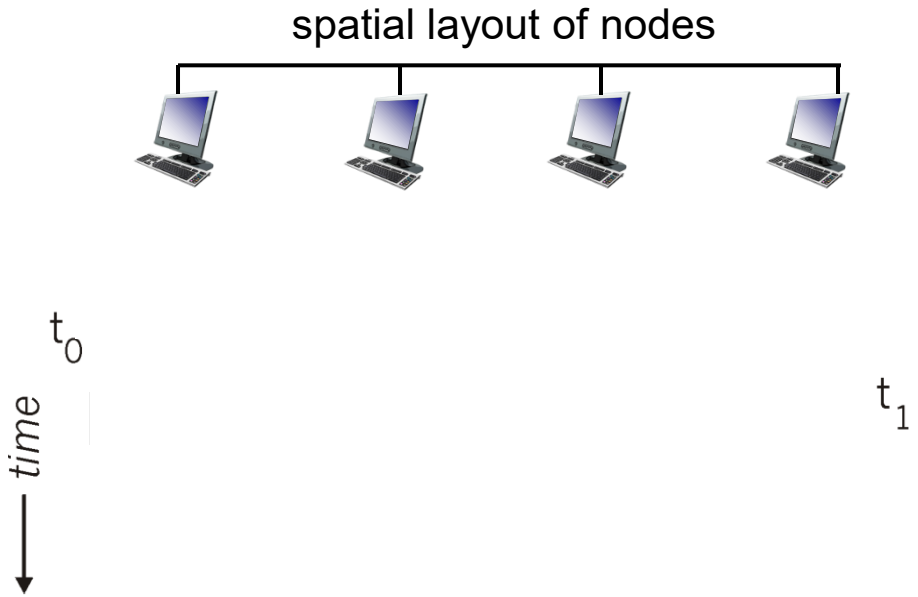
- If channel sensed idle: transmit entire frame
- If channel sensed busy, defer transmission
  - Persistent CSMA: retry immediately with probability p when channel becomes idle (may cause instability)
  - Non-persistent CSMA: retry after random interval
- human analogy: don't interrupt others!

CSMA collisions

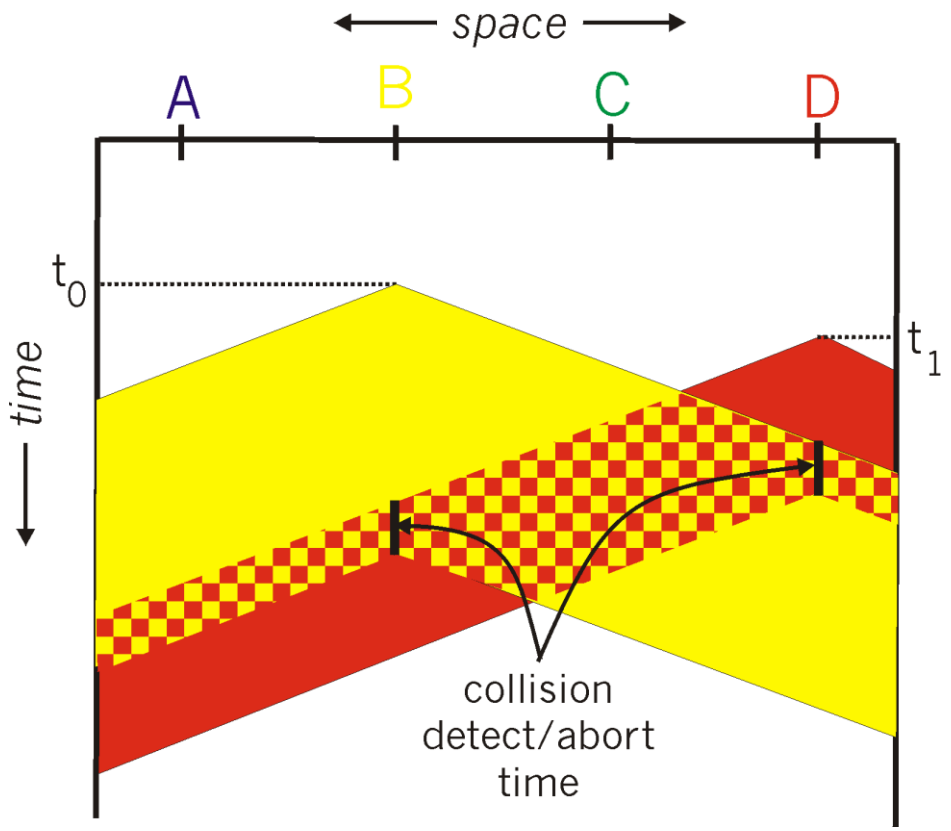
collisions *can* occur:
propagation delay means two nodes may not year hear each other's transmission

collision:
entire packet transmission time wasted

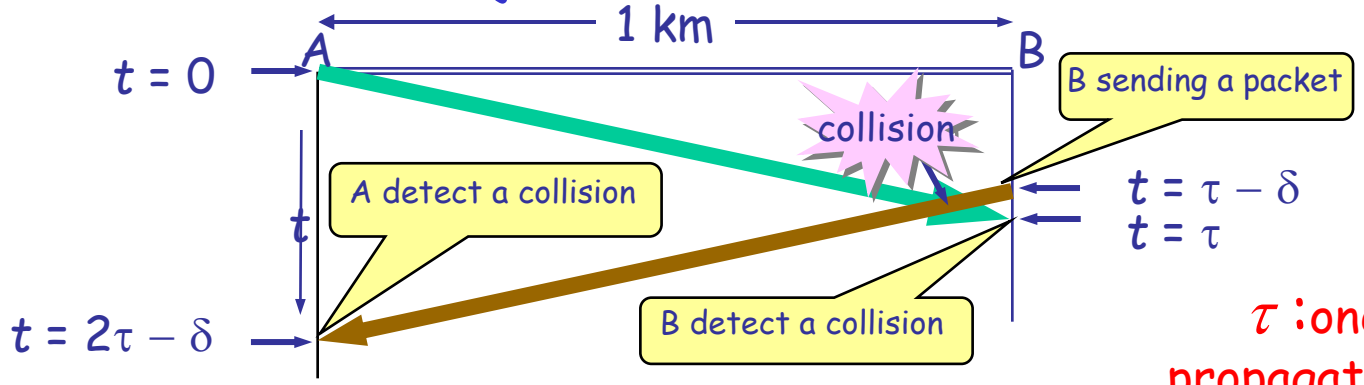
note:
role of distance and propagation delay in determining collision prob.



CSMA/CD collision detection

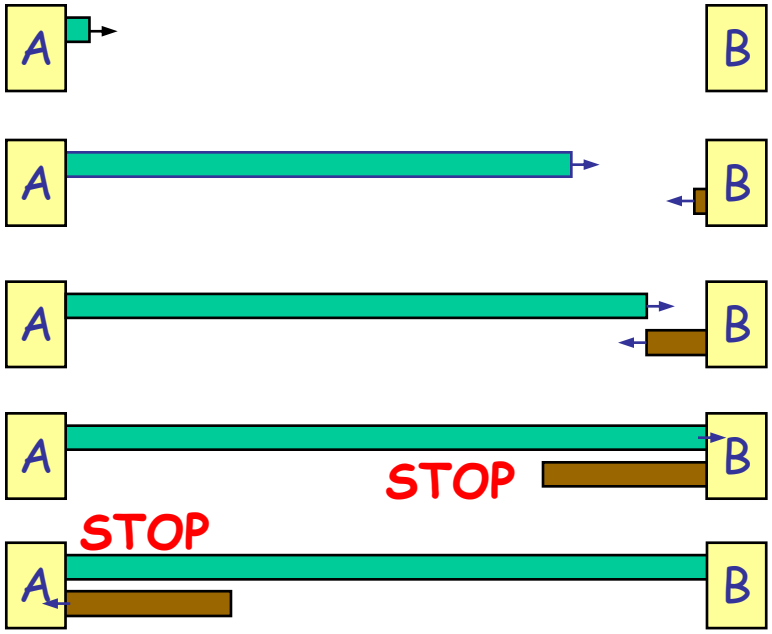


CSMA/CD (Collision Detection)



τ :one way propagation delay

t = 0, A find that the channel is idle, then sending data



t = τ − δ
B find that the channel is idle, then sending data

t = τ − δ / 2
collision

t = τ
B detect the collision, then stop sending

t = 2τ − δ
A detect the collision, then stop sending

2τ :Contention Period

CSMA/CD (Collision Detection)

CSMA/CD: carrier sensing, deferral as in CSMA

- collisions *detected* within short time
- colliding transmissions aborted, reducing channel wastage
- persistent or non-persistent retransmission

- collision detection:
  - easy in wired LANs: measure signal strengths, compare transmitted, received signals
  - difficult in wireless LANs: receiver shut off while transmitting
- human analogy: the polite conversationalist

Summary of MAC protocols

- What do you do with a shared media?
  - Channel Partitioning, by time, frequency or code
    - Time Division,Code Division, Frequency Division
  - Random partitioning (dynamic),
    - ALOHA, S-ALOHA, CSMA, CSMA/CD
    - carrier sensing: easy in some technoligies (wire), hard in others (wireless)
    - CSMA/CD used in Ethernet
  - Taking Turns
    - polling from a central cite, token passing

Summary of MAC protocols

channel partitioning MAC protocols:

- share channel efficiently at high load
- inefficient at low load: delay in channel access, 1/N bandwidth allocated even if only 1 active node!

Random access MAC protocols

- efficient at low load: single node can fully utilize channel
- high load: collision overhead

"taking turns" protocols

look for best of both worlds!

Exercises-4

- A LAN adopts CSMA/CD protocol to realize the media access control. The data transmission rate is 10Mbps, the distance between host A and host B is 2km, and the signal propagation speed is 200000km/s. If there is a conflict when two hosts send data, how long will it take from the time when they start sending data to the time when both hosts detect the conflict? (What is the minimum and maximum duration?)

Chapter 3: Summary

- principles behind data link layer services:
  - framing
  - error detection, correction
  - reliable data transfer
  - sharing a broadcast channel: multiple access
- Next, various link layer technologies
  - LAN Model
  - Ethernet
  - WLAN

Homework

- P319-322
  P9,11,17,22
- P537
  P5